

Vijayi WFH Migration from On-Premises to AWS

Objective

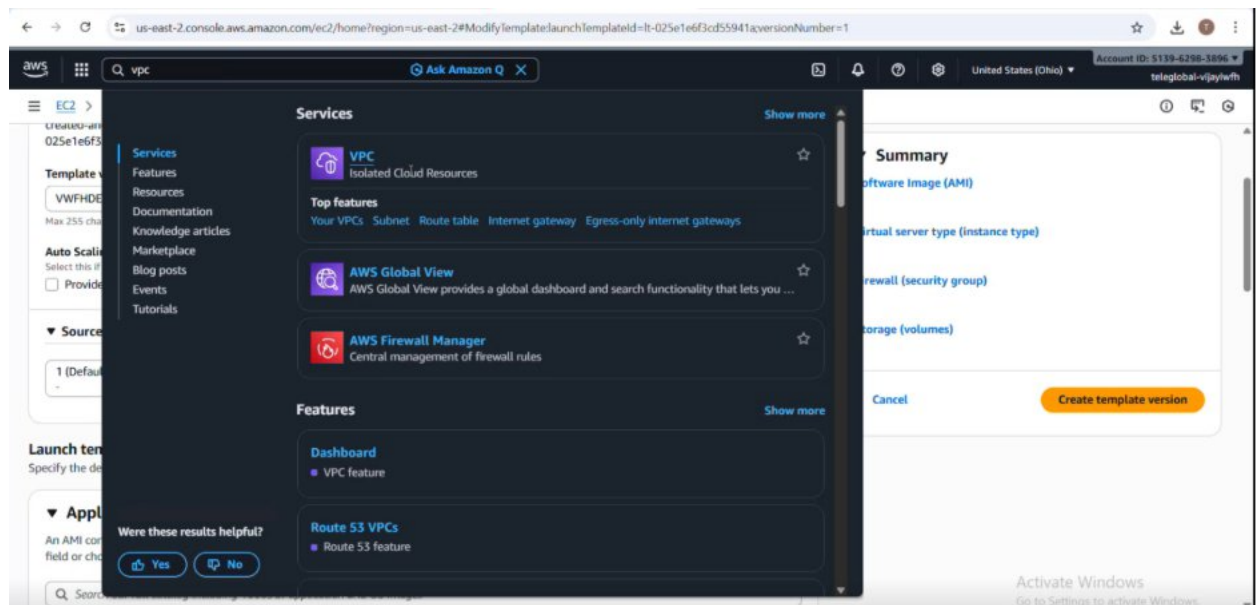
To migrate on-premises servers to AWS using AWS Application Migration Service (MGN).

Prerequisites

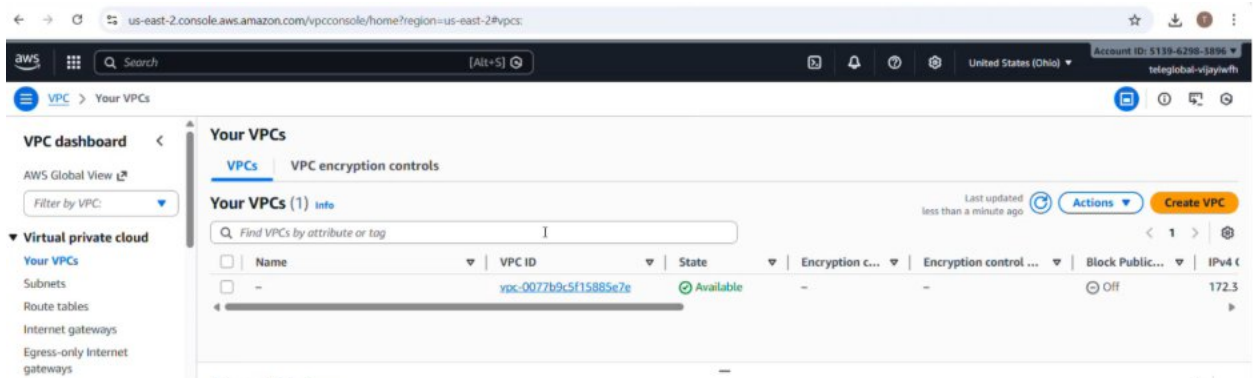
- Backup of on-premises servers
- Admin credentials for on-premises servers
- AWS credentials with MFA
- Approved SOW
- Required AWS services: VPC, IAM, Application Migration Service
- Correct AWS region selection

VPC Creation

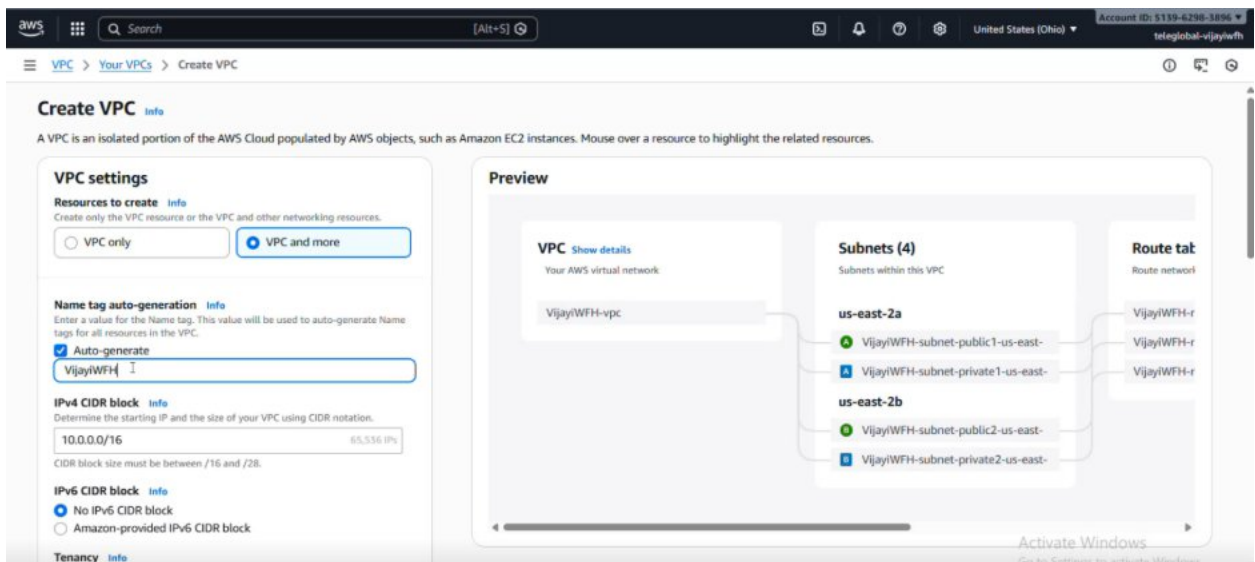
1. Login to AWS Console



2. Open VPC Service — Create VPC



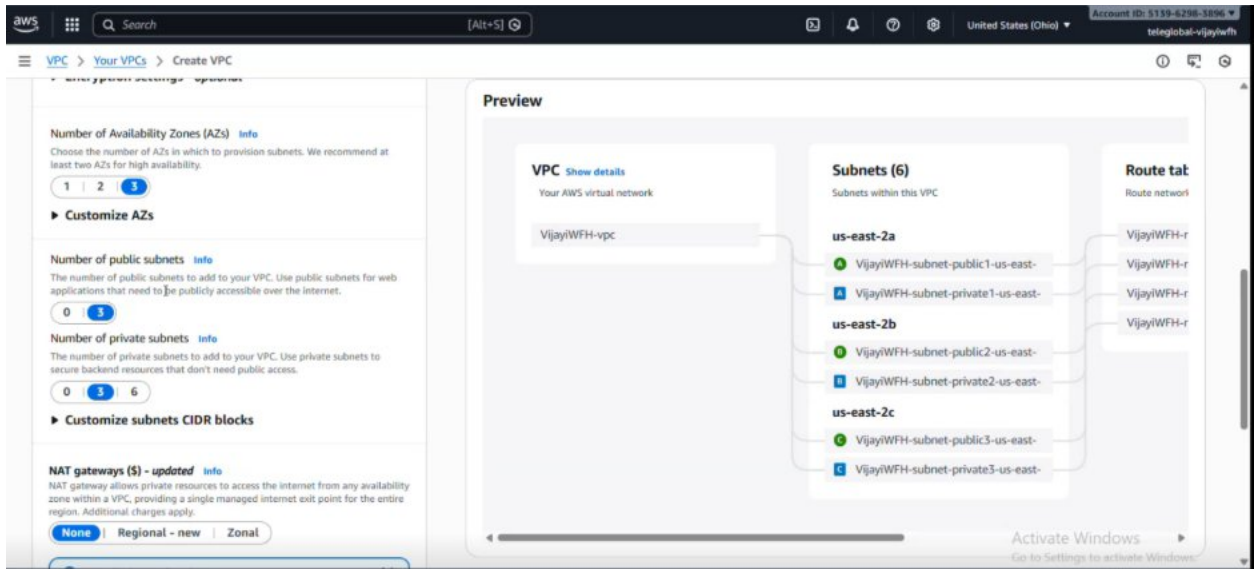
3. Select “VPC and more”



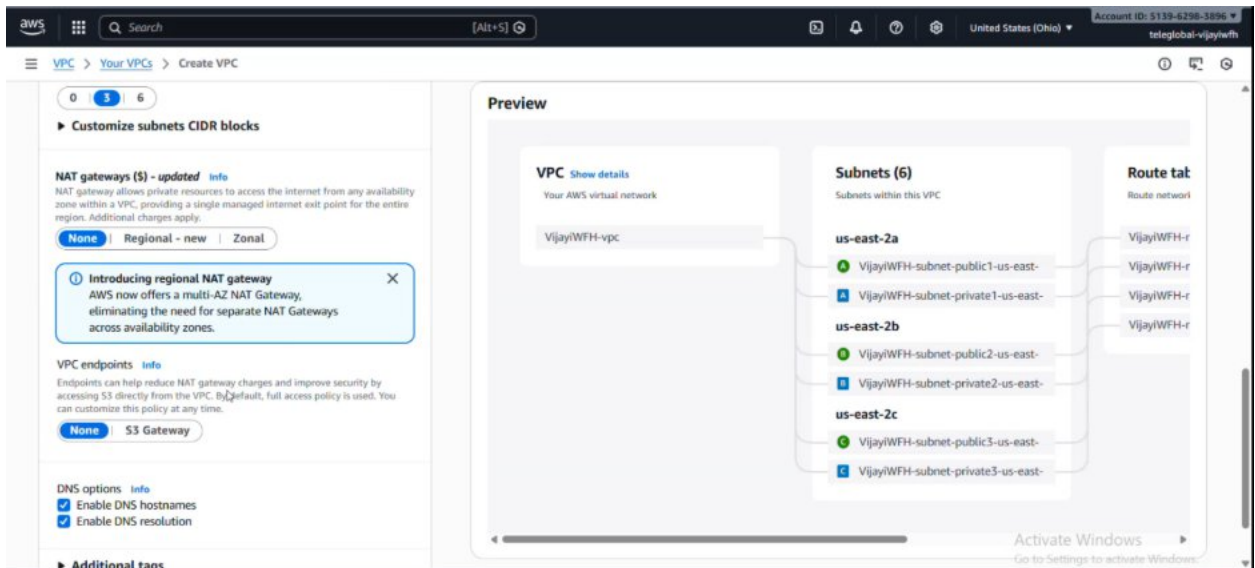
4. VPC Name: VijayiWFH

5. Availability Zones: 3

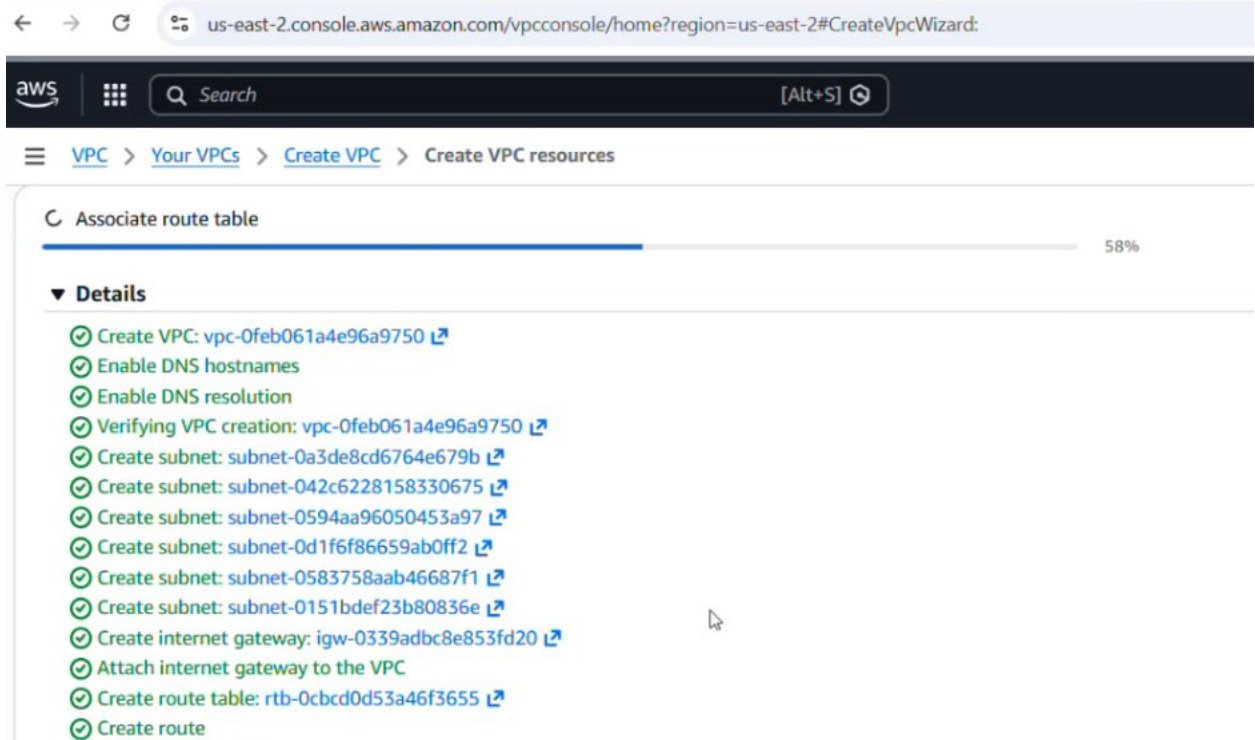
6. Public & Private Subnets: 3 each



7. Enable VPC Endpoints

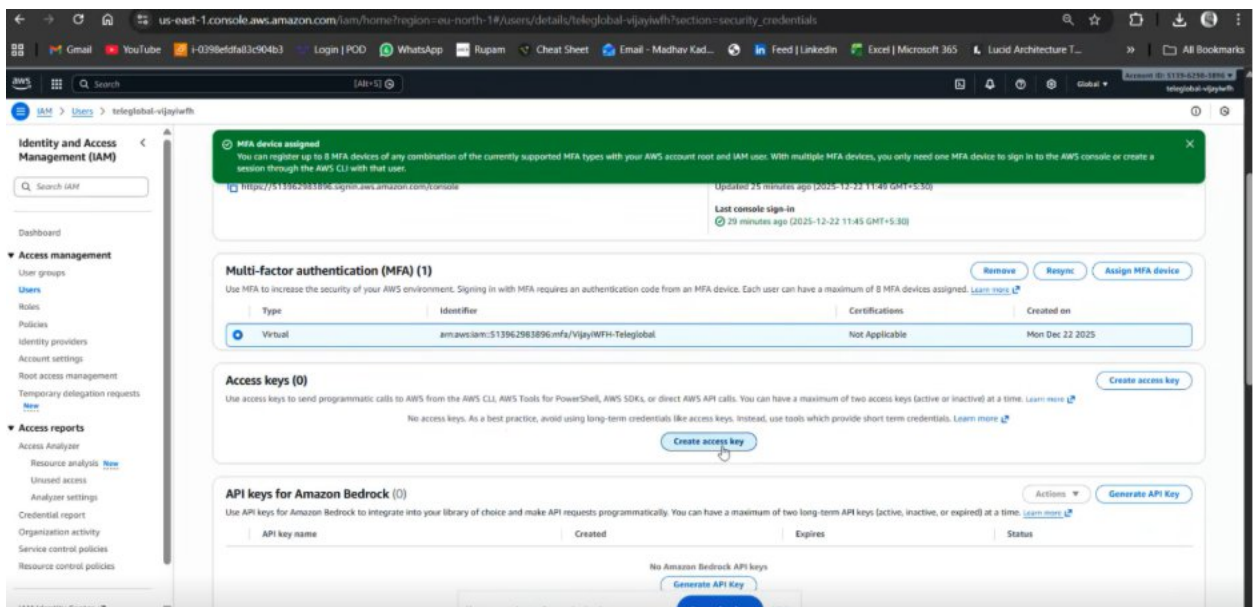


8. Create VPC

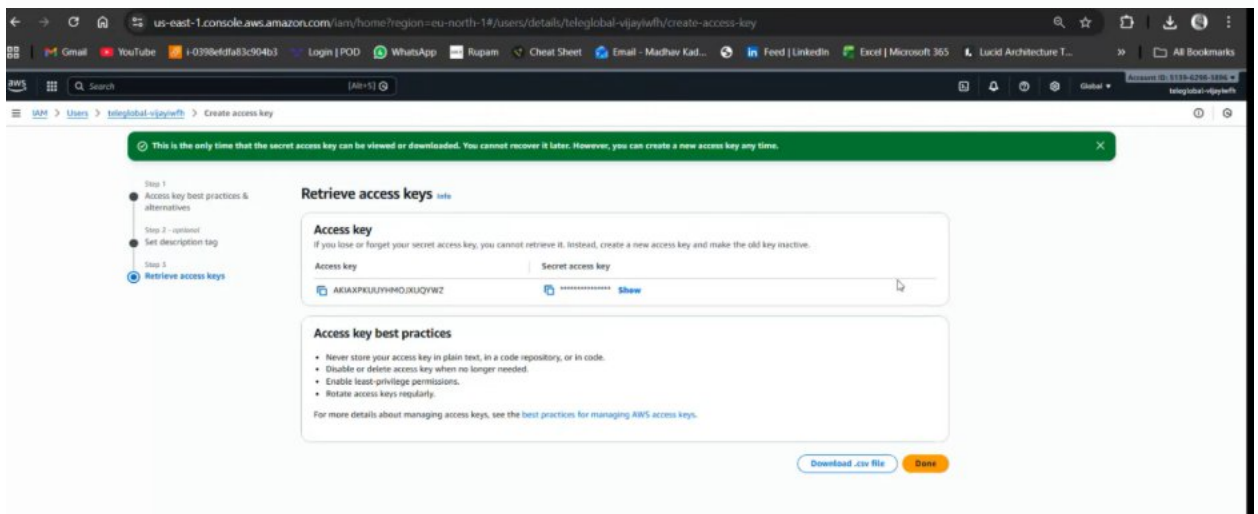
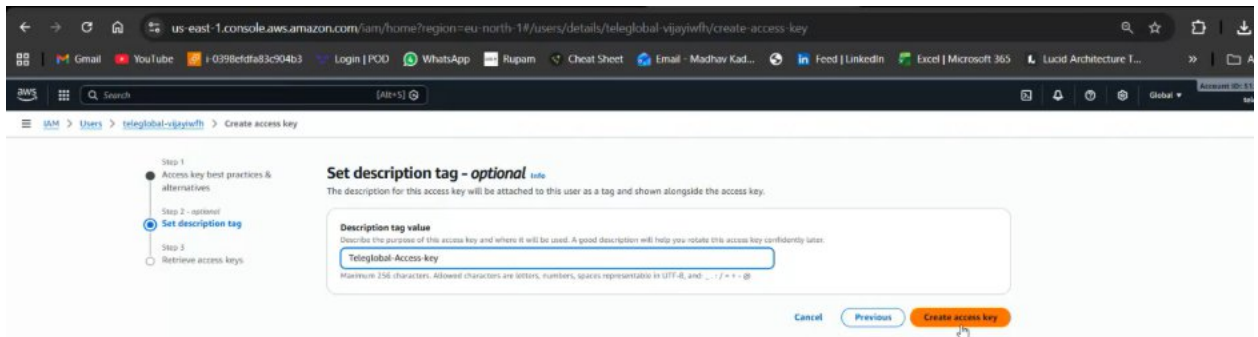
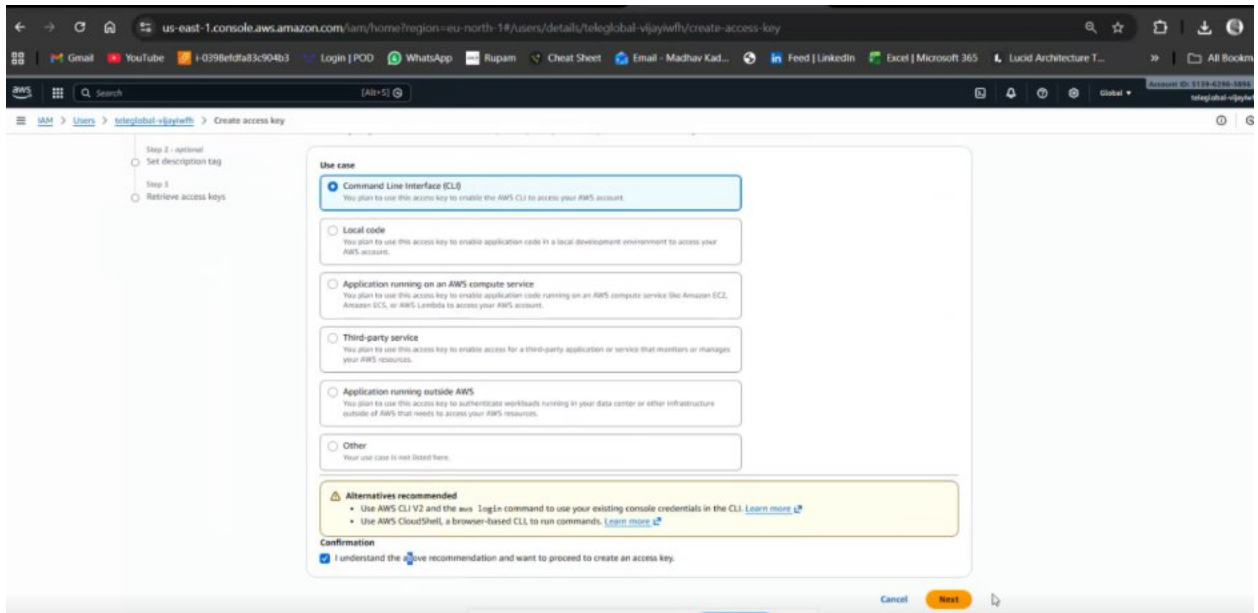


IAM Access Key Generation

1. Go to IAM → Users → Security Credentials



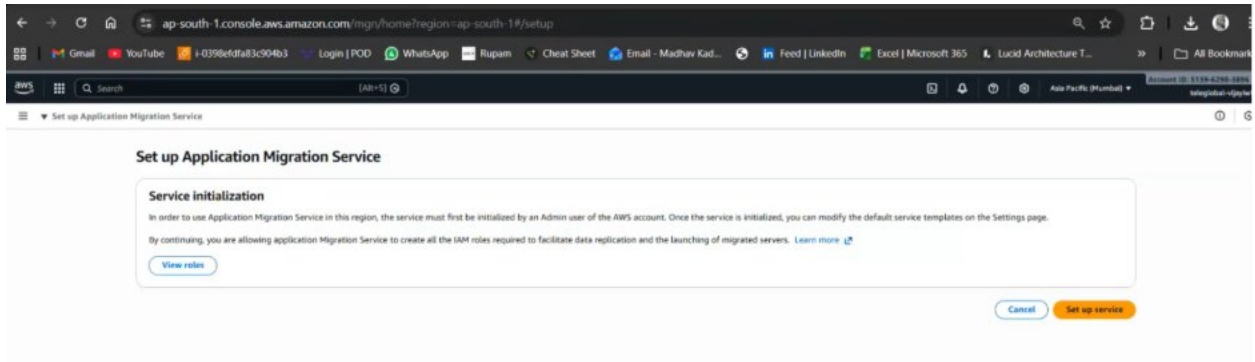
2. Create Access Key (CLI)



3. Download CSV and store securely

Application Migration Service Setup

1. Open Application Migration Service

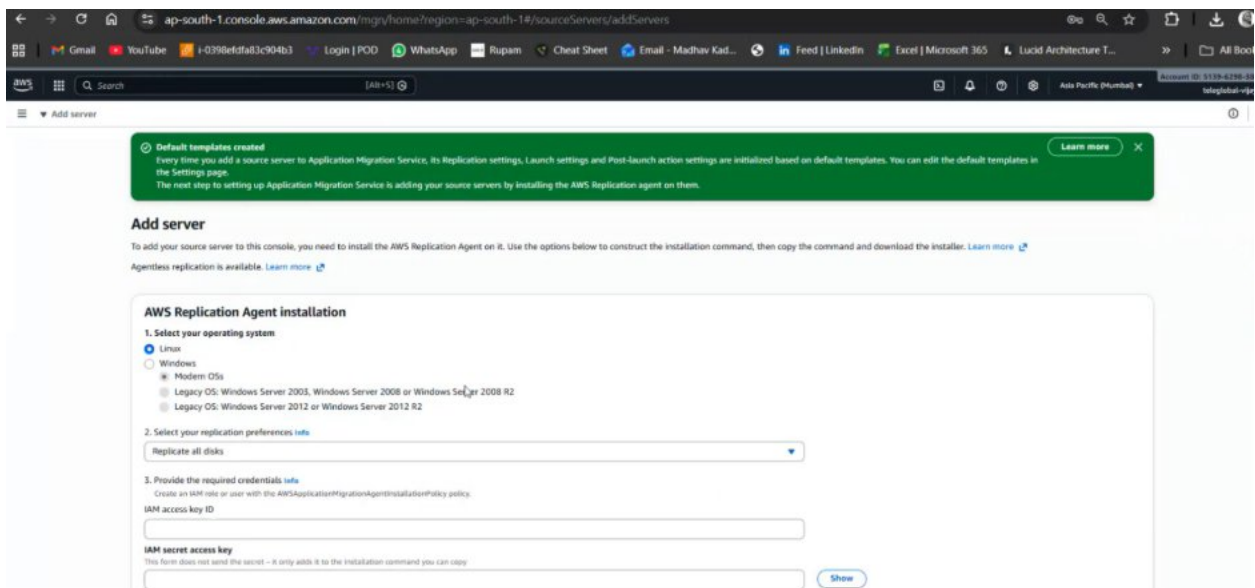


2. Click Set up service

Add Source Servers

1. Source Servers — Add Server

2. Select OS



3. Provide Access & Secret Keys

ap-south-1.console.aws.amazon.com/mgn/home?region=ap-south-1#/sourceServers/addServers

2. Select your replication preferences [info](#)

Replicate all disks

3. Provide the required credentials [info](#)

Create an IAM role or user with the `AWSServiceCatalogAgentInstallationPolicy` policy.

IAM access key ID

AKIAXPKUUYHMQJKUQYVWZ

IAM secret access key

This form does not send the secret - it only adds it to the installation command you can copy

Show

Session token

Session token is only required when using temporary credentials

4. User provided resource id - optional [info](#)

5. Download the installer using this command:

```
sudo wget -O ./aws-replication-installer-init https://aws-application-migration-service-ap-south-1.s3.ap-south-1.amazonaws.com/latest/linux/aws-replication-installer-init.sh
```

If you need to validate the installer hash, the correct hash can be found here:
<https://aws-application-migration-service-hashes-ap-south-1.s3.ap-south-1.amazonaws.com/latest/linux/aws-replication-installer-init.sha512>

6. Copy and input the command below into the command line on your source server

```
sudo chmod +x aws-replication-installer-init; sudo ./aws-replication-installer-init --region ap-south-1 --aws-access-key-id AKIAXPKU
```

Back

4. Download installer and copy commands

5. Download the installer using this command:

```
sudo wget -O ./aws-replication-installer-init https://aws-application-migration-service-ap-south-1.s3.ap-south-1.amazonaws.com/latest/linux/aws-replication-installer-init.sh
```

If you need to validate the installer hash, the correct hash can be found here:
<https://aws-application-migration-service-hashes-ap-south-1.s3.ap-south-1.amazonaws.com/latest/linux/aws-replication-installer-init.sha512>

6. Copy and input the command below into the command line on your source server

```
sudo chmod +x aws-replication-installer-init; sudo ./aws-replication-installer-init --region ap-south-1 --aws-access-key-id AKIAXPKU
```

On-Prem Server Setup

1. Login to server

2. sudo su

teleglobaluser@deckard1209: ~

login as: teleglobaluser

teleglobaluser@50.30.32.116's password:

Dear customer,

If you want to use your own kernel, please make sure you don't touch the kernel boot parameters (append) as some of our hardware requires the parameters `acpi=ht` and/or `noapic`.

The programs included with the Ubuntu system are free software; the exact distribution terms for each program are described in the individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.

To run a command as administrator (user "root"), use "`sudo <command>`". See "`man sudo_root`" for details.

teleglobaluser@deckard1209:~\$ sudo su
[sudo] password for teleglobaluser: █

3. Run installer commands

```
teleglobaluser@deckard1209:~$ login as: teleglobaluser
teleglobaluser@950.30.32.116's password:

Dear customer,

If you want to use your own kernel, please make sure you don't touch the
kernel boot parameters (append) as some of our hardware requires the
parameters acpi=ht and/or noapic.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

teleglobaluser@deckard1209:~$ sudo su
[sudo] password for teleglobaluser:
Sorry, try again.
[sudo] password for teleglobaluser:
deckard1209:/home/teleglobaluser# http
deckard1209:/home/teleglobaluser# sudo wget -O ./aws-replication-installer-init https://aws-application-migration-service-ap-south-1.s3.ap-south-1.amazonaws.com/latest/linux/aws-replication
-installer-init
I

teleglobaluser@deckard1209:~$ login as: teleglobaluser
teleglobaluser@950.30.32.116's password:

Dear customer,

If you want to use your own kernel, please make sure you don't touch the
kernel boot parameters (append) as some of our hardware requires the
parameters acpi=ht and/or noapic.

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individual files in /usr/share/doc/*/copyright.

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applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

teleglobaluser@deckard1209:~$ sudo su
[sudo] password for teleglobaluser:
Sorry, try again.
[sudo] password for teleglobaluser:
deckard1209:/home/teleglobaluser# http
deckard1209:/home/teleglobaluser# sudo wget -O ./aws-replication-installer-init https://aws-application-migration-service-ap-south-1.s3.ap-south-1.amazonaws.com/latest/linux/aws-replication
-installer-init
--2025-12-22 09:59:36-- https://aws-application-migration-service-ap-south-1.s3.ap-south-1.amazonaws.com/latest/linux/aws-replication-installer-init
Resolving aws-application-migration-service-ap-south-1.s3.ap-south-1.amazonaws.com (aws-application-migration-service-ap-south-1.s3.ap-south-1.amazonaws.com)... 52.219.64.11, 3.5.211.242, 5
2.219.62.7, ...
Connecting to aws-application-migration-service-ap-south-1.s3.ap-south-1.amazonaws.com (aws-application-migration-service-ap-south-1.s3.ap-south-1.amazonaws.com)|52.219.64.11|:443... connec
ted.
HTTP request sent, awaiting response... 200 OK
Length: 22824816 (22M) [Binary/octet-stream]
Saving to: './aws-replication-installer-init'

./aws-replication-installer-init 100%[=====] 21.77M 5.94MB/s in 3.7s

2025-12-22 09:59:41 (5.94 MB/s) - './aws-replication-installer-init' saved [22824816/22824816]

deckard1209:/home/teleglobaluser# sudo chmod +x aws-replication-installer-init; sudo ./aws-replication-installer-init --region ap-south-1 --aws-access-key-id AKIAJFPUUTDMDJXUQTWE --aws-secre
t-access-key oF4IXTFBjOCWnIdJgd/dgXPh3PHTAo36Se8+dr2m --no-prompt
```

```
teleglobaluser@deckard1209:~$ sudo su
[sudo] password for teleglobaluser:
teleglobaluser@deckard1209:~$ sudo su
[sudo] password for teleglobaluser:
teleglobaluser@deckard1209:~$ http
2025-12-22 09:59:36-- https://aws-application-migration-service-ap-south-1.s3.ap-south-1.amazonaws.com/latest/linux/aws-replication
Resolving aws-application-migration-service-ap-south-1.s3.ap-south-1.amazonaws.com (aws-application-migration-service-ap-south-1.s3.ap-south-1.amazonaws.com)... 52.219.64.11, 3.5.211.242, 5
2.219.62.7, ...
Connecting to aws-application-migration-service-ap-south-1.s3.ap-south-1.amazonaws.com (aws-application-migration-service-ap-south-1.s3.ap-south-1.amazonaws.com) (52.219.64.11):443... connec
ted.
HTTP request sent, awaiting response... 200 OK
Length: 22824816 (22M) [binary/octet-stream]
Saving to: './aws-replication-installer-init'

./aws-replication-installer-init 100%[=====] 21.77M 5.94MB/s in 3.7s

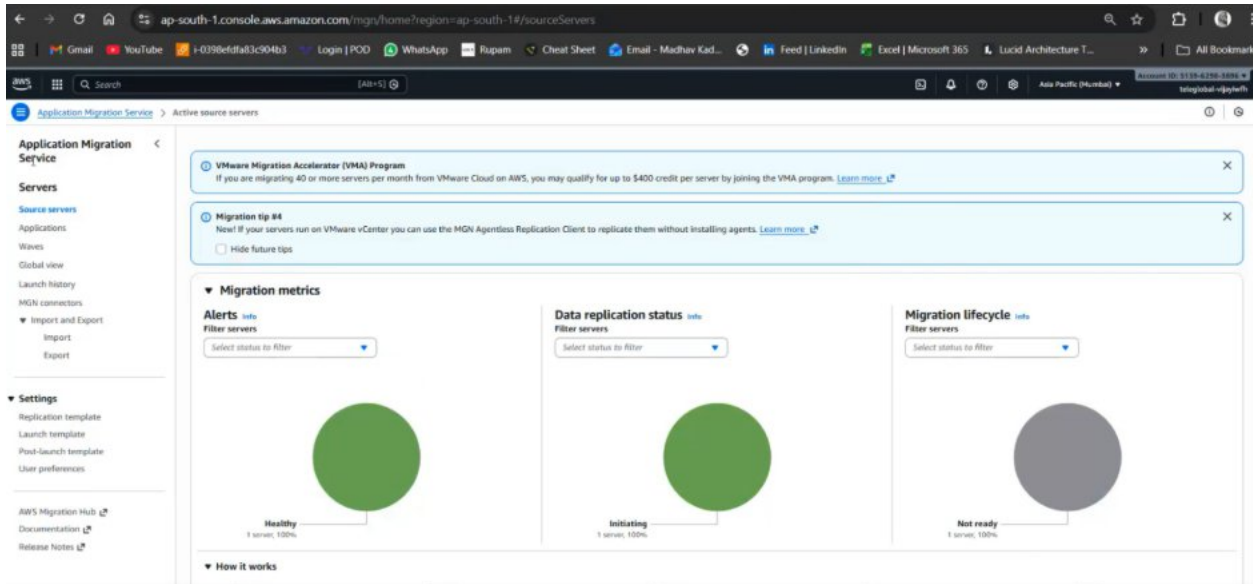
2025-12-22 09:59:41 (5.94 MB/s) - './aws-replication-installer-init' saved [22824816/22824816]

teleglobaluser@deckard1209:~$ sudo chmod +x aws-replication-installer-init; sudo ./aws-replication-installer-init --region ap-south-1 --aws-access-key-id AKIAJFKUUYHMOJXUQWVE --aws-sect
et-access-key 094IXIFEJocKndJgd/daxRh3FNEAo365e8*dr3e --no-prompt
The installation of the AWS Replication Agent has started.
Identifying volumes for replication.
Identified volume for replication: /dev/sda of size 1864 GiB
Identified volume for replication: /dev/sdb of size 1864 GiB
All volumes for replication were successfully identified.
Downloading the AWS Replication Agent onto the source server...
Finished.
Installing the AWS Replication Agent onto the source server...
Finished.
Syncing the source server with the Application Migration Service Console...
Finished.
The following is the source server ID: s-e79b3beadbe01882.
You now have 1 active source server out of a total quota of 150.
Learn more about increasing source servers limit at https://docs.aws.amazon.com/mgn/latest/ug/MGN-service-limits.html
The AWS Replication Agent was successfully installed.
teleglobaluser@deckard1209:~$
```

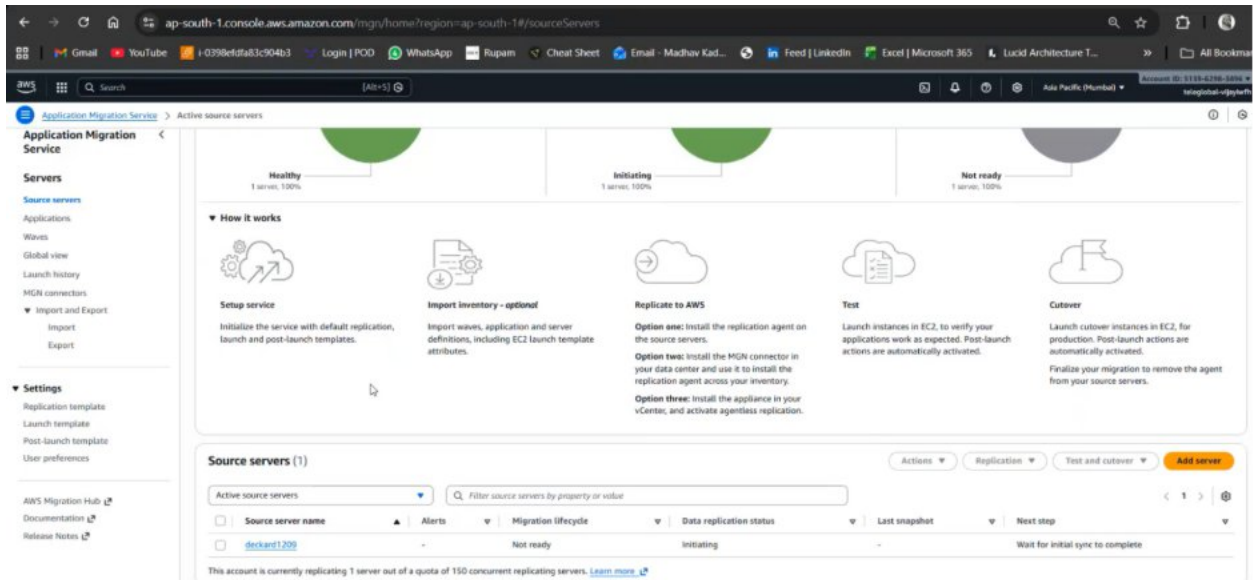
4. Repeat for all servers

Replication Monitoring

- Monitor status under Source Servers

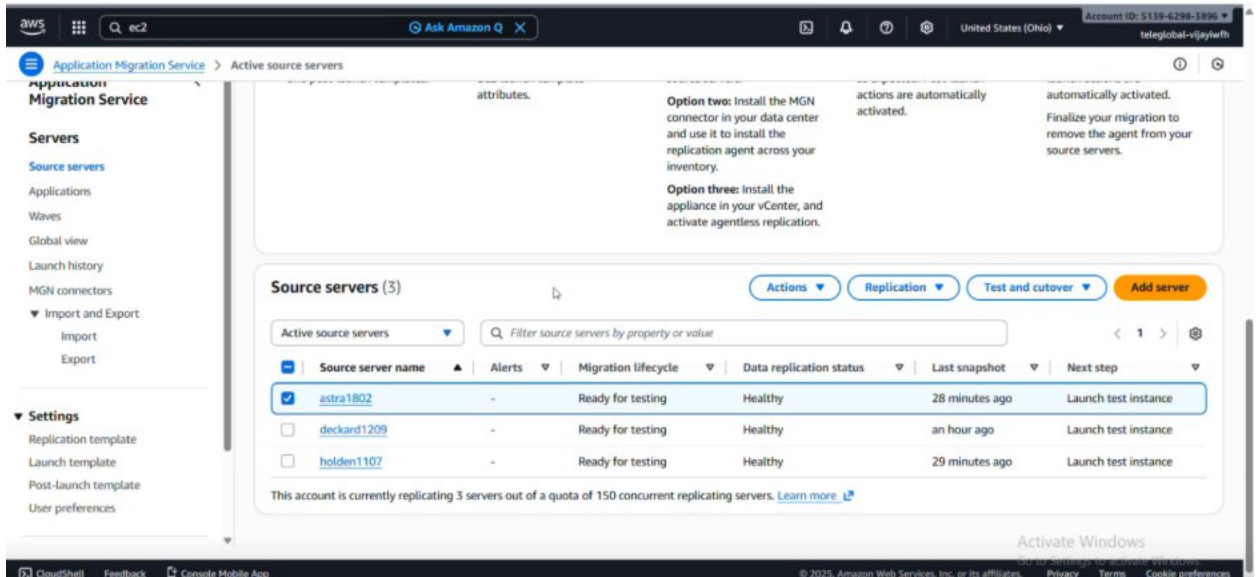


- Status changes to Ready for Testing



Launch Template Configuration

- Identify Launch Template ID



Employee-Evaluation-Form-PC Launch template | AWS Appli Launch templates | EC2 | us-east-2

us-east-2.console.aws.amazon.com/mgn/home?region=us-east-2#/launchTemplate

Launch template

Application Migration Service

Servers

Source servers

Applications

Waves

Global view

Launch history

MGN connectors

▼ Import and Export

Import

Export

Settings

Replication template

Launch template

Post-launch template

User preferences

General launch settings

Configure the default settings that will be used when launching target servers.

Instance type right sizing

On

Copy private IPv4 address

No

Transfer server tags

No

Start instance upon launch

Yes

Operating system licensing

License-included

Boot mode

Use source

Default EC2 Launch Template

Configure the default settings that will be applied to the EC2 launch template of every target server.

Template ID

lt-025e1e6f3cd55941a

Instance type (ignored when right-sizing is active)

-

Security groups

-

Default target subnet

-

EBS volume type

General Purpose SSD (gp3)

MAP program tagging

Configure MAP resource tags to be applied to all instances launched by this service.

Activate Windows

Go to Settings to activate Windows.

Search [Alt+S] Ask Amazon Q

Launch Templates (4)

Search

☐	Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time	Created By
☐	lt-0d3fa4e694ca7a8ca	created-and-used-by-application...	2	2	2025-12-22T11:58:00.000Z	arn:aws:sts:5139625
☐	lt-025e1e6f3cd55941a	created-and-used-by-application...	1	1	2025-12-22T11:37:45.000Z	arn:aws:iam:513962
☐	lt-087bb4cc4cd578f19	created-and-used-by-application...	2	2	2025-12-22T11:44:05.000Z	arn:aws:sts:5139625
☐	lt-0c7e3af18d7b7669b	created-and-used-by-application...	2	2	2025-12-23T07:39:21.000Z	arn:aws:sts:5139625

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#ModifyTemplate:launchTemplateId=lt-025e1e6f3cd55941a;versionNumber=1

EC2 > Launch templates > Modify template (Create new version)

Modify template (Create new version)

Modifying a template allows you to create a new template version from an existing version. Using versions allows you to manage templates in a structured and controlled way. It also allows you to always use the default version of the template by rolling out updates to templates without having to change a reference to the template name or ID.

Launch template name and version description

Launch template name

created-and-used-by-application-migration-service-account_level_template-252212-113745 (lt-025e1e6f3cd55941a)

Template version description

A prod webserver for MyApp

Max 255 chars

Auto Scaling guidance

Select this if you intend to use this template with EC2 Auto Scaling

☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

Source template

1 (Default)

Summary

Software Image (AMI)

-

Virtual server type (instance type)

-

Firewall (security group)

-

Storage (volumes)

-

Cancel

Create template version

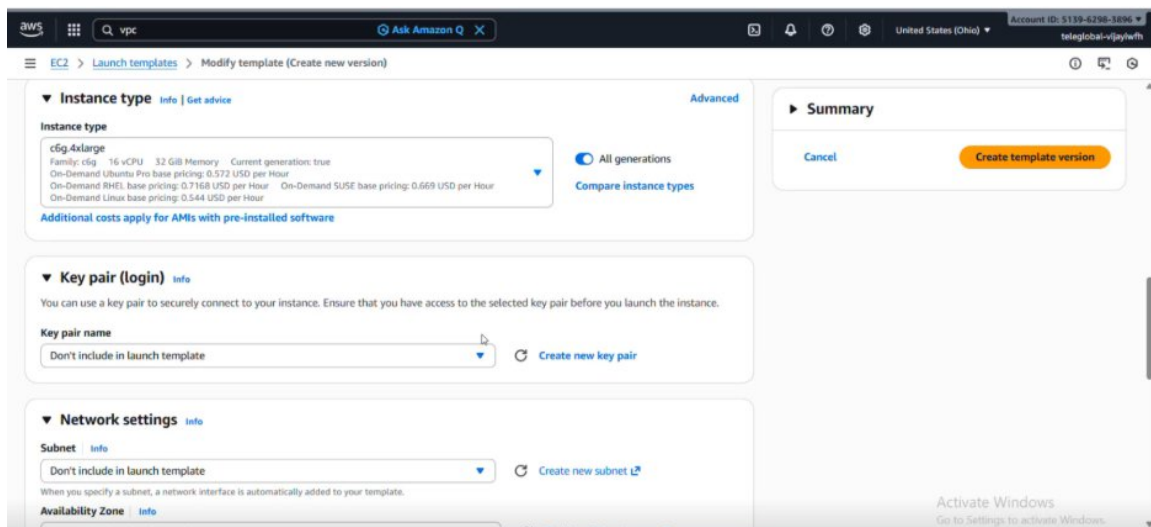
Activate Windows

Go to Settings to activate Windows.

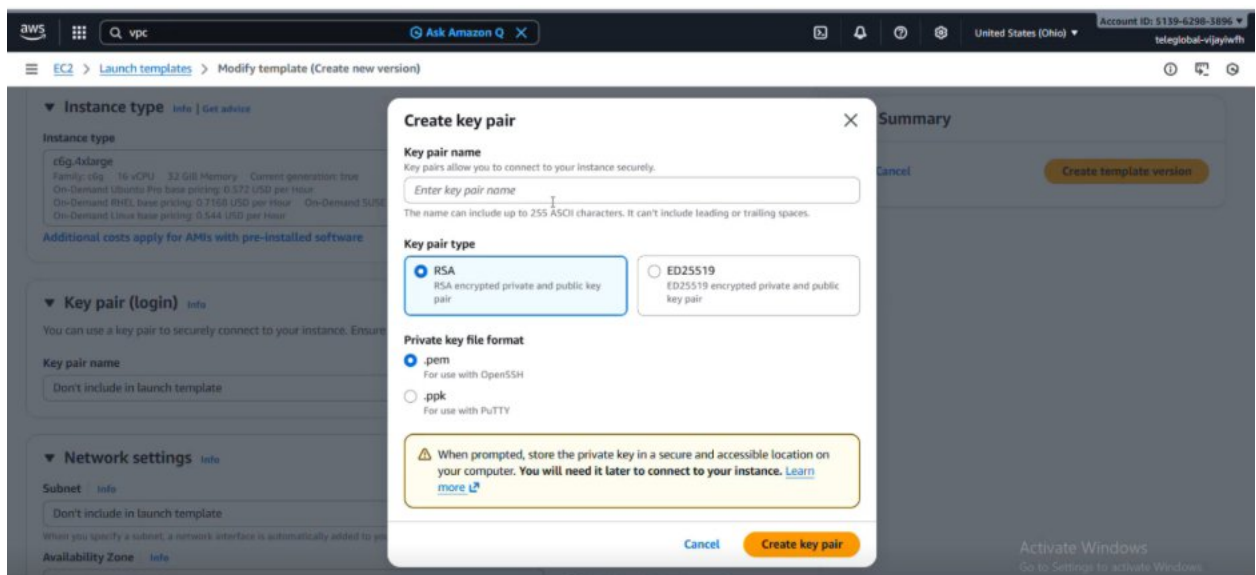
CloudShell Feedback Console Mobile App

© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

- Select instance type as per BOQ



- Create and assign Key Pair



- Configure Security Group (SSH, HTTP, HTTPS)

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#ModifyTemplate:launchTemplateId=lt-025e1e6f3cd55941a&versionNumber=1

EC2 > Launch templates > Modify template (Create new version)

Add security group rule

Advanced network configuration

Network interface 1

Device index 0

Subnet subnet-0a3de8cd6764e679b

Primary IP 123.123.123.1

IPv4 Prefixes

Delete on termination

Network interface New interface

Security groups New security group

Secondary IP Don't include in launch template

IPv6 Prefixes

Interface type

Description

Auto-assign public IP Enable

IPv6 IPs

Assign Primary IPv6 IP

Network card index

Summary

Cancel

Create template version

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#ModifyTemplate:launchTemplateId=lt-025e1e6f3cd55941a&versionNumber=1

EC2 > Launch templates > Modify template (Create new version)

CPU options - update

Use default CPU options

Default active vCPUs 16

AMD SEV-SNP

Metadata accessible

Summary

Cancel

Create template version

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#ModifyTemplate:launchTemplateId=lt-025e1e6f3cd55941a&versionNumber=1

EC2 > Launch templates > Modify template (Create new version)

Success

Successfully created created-and-used-by-application-migration-service-account_level_template-252212-113745(lt-025e1e6f3cd55941a).

Actions log

Next Steps

Launch an instance

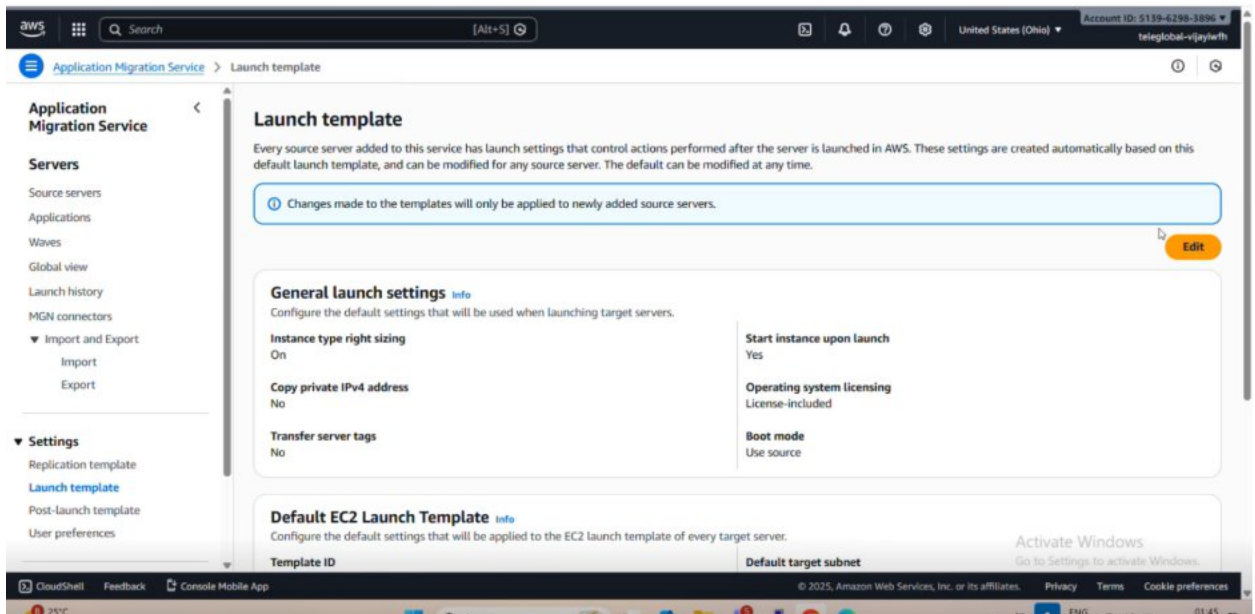
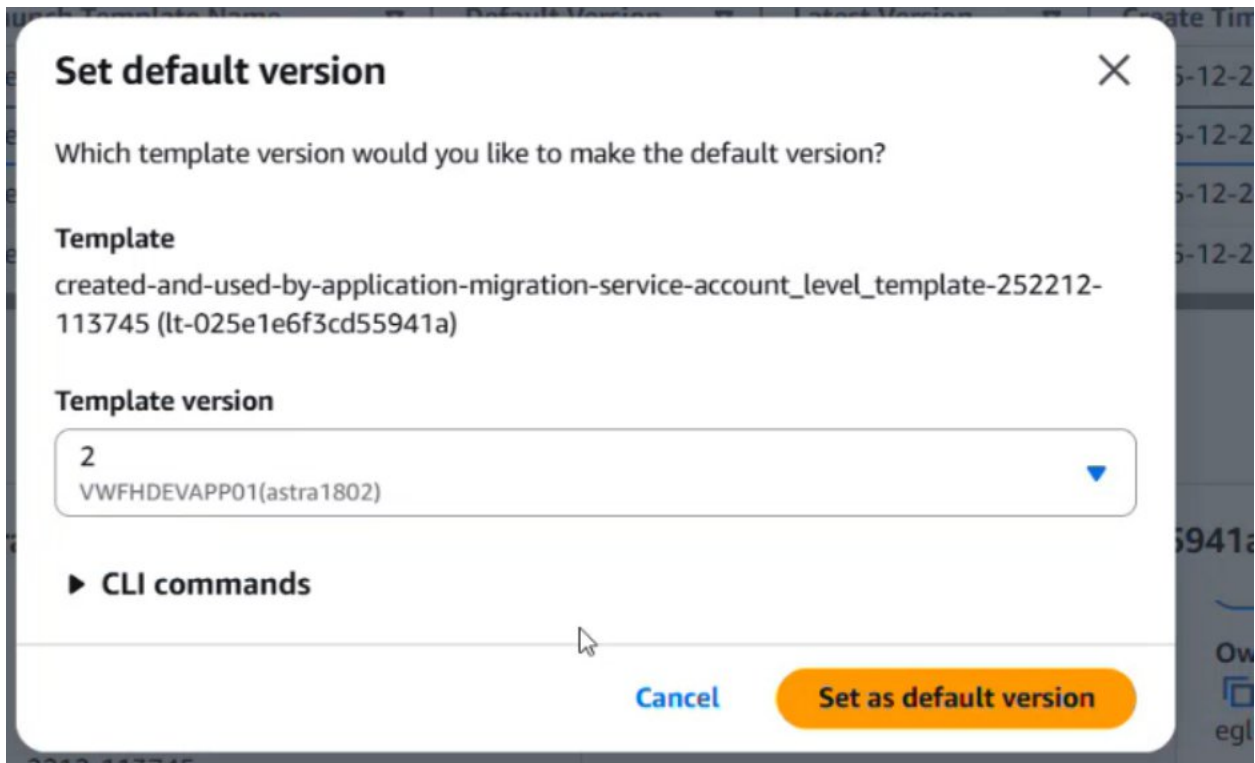
Create an Auto Scaling group from your template

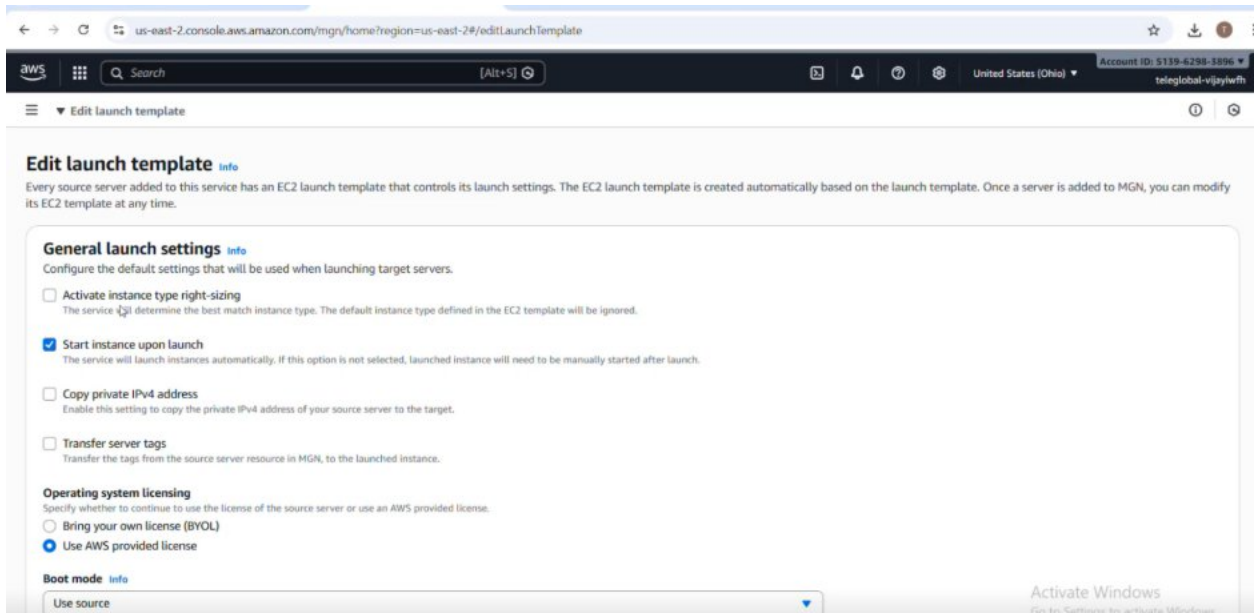
Create Spot Fleet

Summary

Cancel

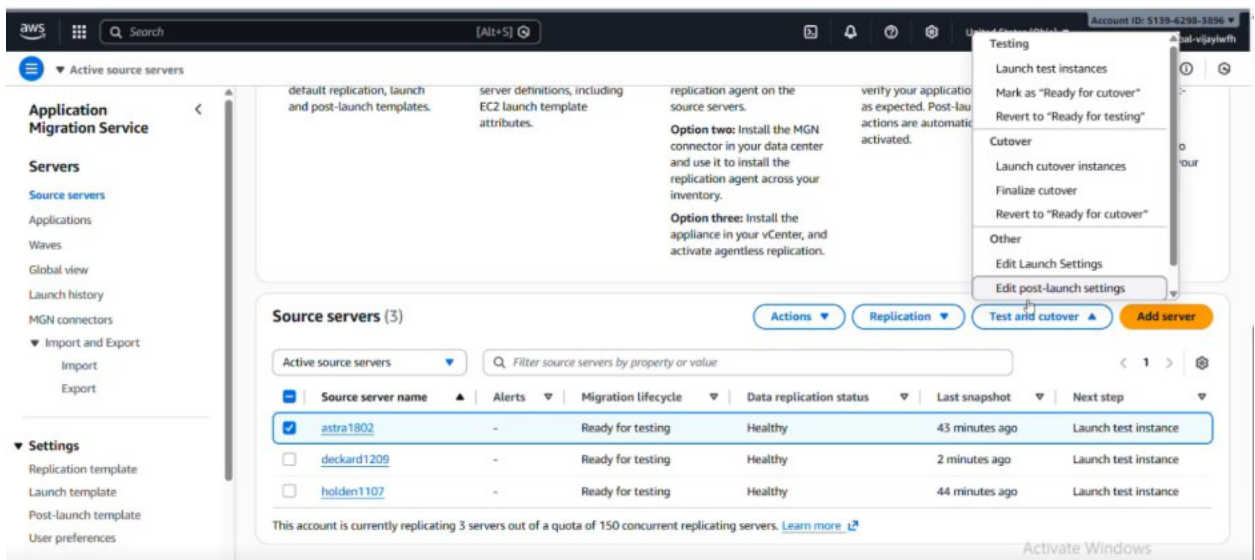
Create template version





Test Instance Launch

- Test & Cutover → Launch Test Instances



Launch test instance for 1 server

You are about to launch EC2 instance for 1 server.

These instances will be launched according to the Launch Settings you have configured for them. Launched instances accrue EC2 charges as per your AWS account's rates. [Learn more](#)

▼ The action will be applied to the following servers

astra1802

Cancel

Launch

• Monitor EC2 instance

Instances (3) info

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
AWS Application Migration Service Replication Server	i-04dddb030b6c31582	Running	m6a.xlarge	3/3 checks passed	View alarms +	us-east-2
AWS Application Migration Service Replication Server	i-02e07cdeca5709323	Running	m6a.2xlarge	3/3 checks passed	View alarms +	us-east-2
AWS Application Migration Service Conversion Server	i-0cb0850b2e7ecb684	Initializing	m5.large	1/3 checks passed	View alarms +	us-east-2

Instances (3) info

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
AWS Application Migration Service Replication Server	i-04dddb030b6c31582	Running	m6a.xlarge	3/3 checks passed	View alarms +	us-east-2
AWS Application Migration Service Replication Server	i-02e07cdeca5709323	Running	m6a.2xlarge	3/3 checks passed	View alarms +	us-east-2
AWS Application Migration Service Conversion Server	i-0cb0850b2e7ecb684	Stopping	m5.large	1/3 checks passed	View alarms +	us-east-2

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
astra1802	i-0cdea19f02fd3ae9c	Running	c6a.4xlarge	3/3 checks passed	View alarms +	us-east-1
AWS Application Migration Service Replication Server	i-04ddd030b6c31582	Running	m6a.xlarge	3/3 checks passed	View alarms +	us-east-1
AWS Application Migration Service Replication Server	i-02e07cdea5709323	Running	m6a.2xlarge	3/3 checks passed	View alarms +	us-east-1
AWS Application Migration Service Conversion Server	i-04635c69b8b0b25a6	Terminated	m5.large	-	View alarms +	us-east-1
AWS Application Migration Service Conversion Server	i-06f9ef1b20f005a54	Terminated	m5.large	-	View alarms +	us-east-1
AWS Application Migration Service Conversion Server	i-0ef7e87e201f237c3	Terminated	m5.large	-	View alarms +	us-east-1
astra1802	i-0a26bf48484ed835c	Terminated	m5.2xlarge	-	View alarms +	us-east-1
astra1802	i-008465e24470207b0	Terminated	m5.2xlarge	-	View alarms +	us-east-1

Troubleshooting

- Check Job ID status

General launch settings

Instance type right sizing: On	Operating system licensing: BYOL	MAP program tagging Add MAP tag to launched instance: No MAP tag value: -
Start instance upon launch: Yes	Transfer server tags: No	
Copy private IPv4 address: No	Boot mode: Legacy BIOS	

EC2 Launch Template

Template ID: lt-0c7e3af18d7b7669b	Primary network interface: Description
Instance type (ignored when right-sizing is active): m5.2xlarge	Subnet: -

The first screenshot shows the 'Migration dashboard' for a migration job. The lifecycle progress bar indicates the job is 'Ready for testing'. The 'Launch status' is '-', and the 'Last test' job ID is 'mgmjob-48d770853f4fe587f'.

The second screenshot shows the 'EC2 Launch Template' configuration. The template ID is 'lt-0c7e3af18d7b7669b'. The instance type is 'm5.2xlarge'. The EBS volumes are 'Volume 1 (233 GiB, EBS, General Purpose SSD (gp3))' and 'Volume 2 (233 GiB, EBS, General Purpose SSD (gp3))'. The primary network interface is 's-40b6c012acd2f5b7e'.

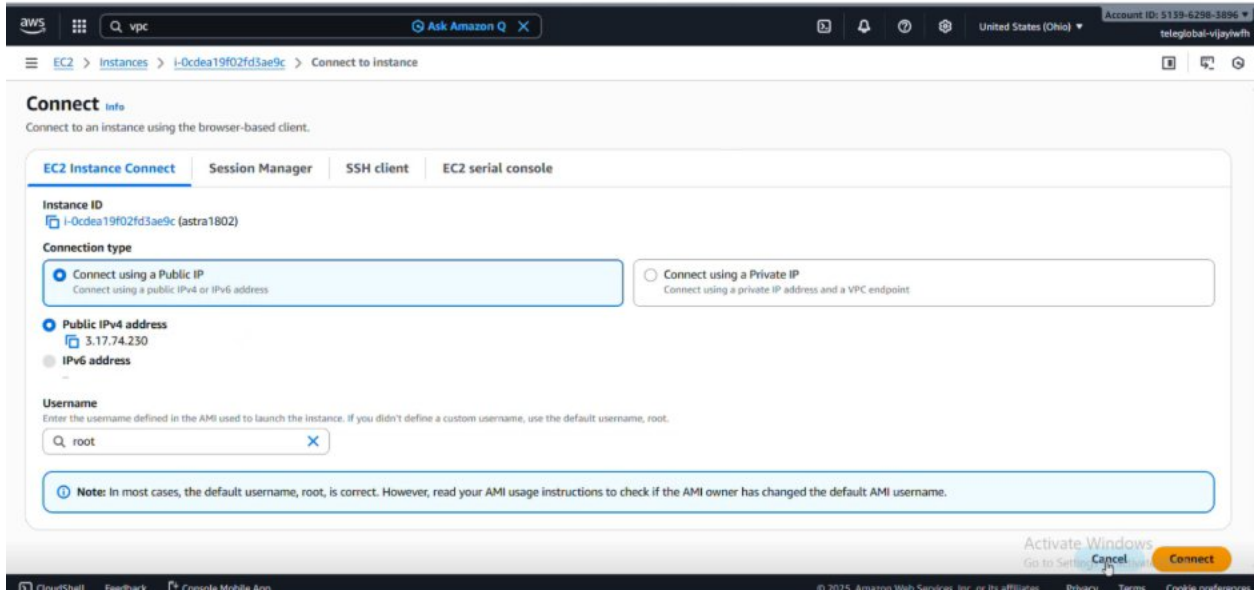
The third screenshot shows the 'Job history' for the migration job. The job ID is 'mgmjob-48d770853f4fe587f'. The history shows the following steps:

Timestamp	Event	Source server
26 December 2025 at 02:29 (UTC-8:00)	Started taking snapshot	s-40b6c012acd2f5b7e
26 December 2025 at 02:29 (UTC-8:00)	Finished taking snapshot	s-40b6c012acd2f5b7e
26 December 2025 at 02:29 (UTC-8:00)	Conversion started	s-40b6c012acd2f5b7e
26 December 2025 at 02:34 (UTC-8:00)	Conversion succeeded	s-40b6c012acd2f5b7e
26 December 2025 at 02:34 (UTC-8:00)	Started launching test/ cutover EC2 instance	s-40b6c012acd2f5b7e
26 December 2025 at 02:34 (UTC-8:00)	Launching of test/ cutover instance failed	s-40b6c012acd2f5b7e

The error message for the failed launch is: 'Error: An error occurred (InvalidParameterValue) when calling the RunInstances operation: InvalidParameterValue: The provided value for the parameter 'Subnet' is not valid. The value must be a valid Subnet ID or Subnet ARN.'

Post Migration Validation

- Verify SSH connectivity
- Validate application services



Completion

=====

Repeat for all servers

=====

Open VPN

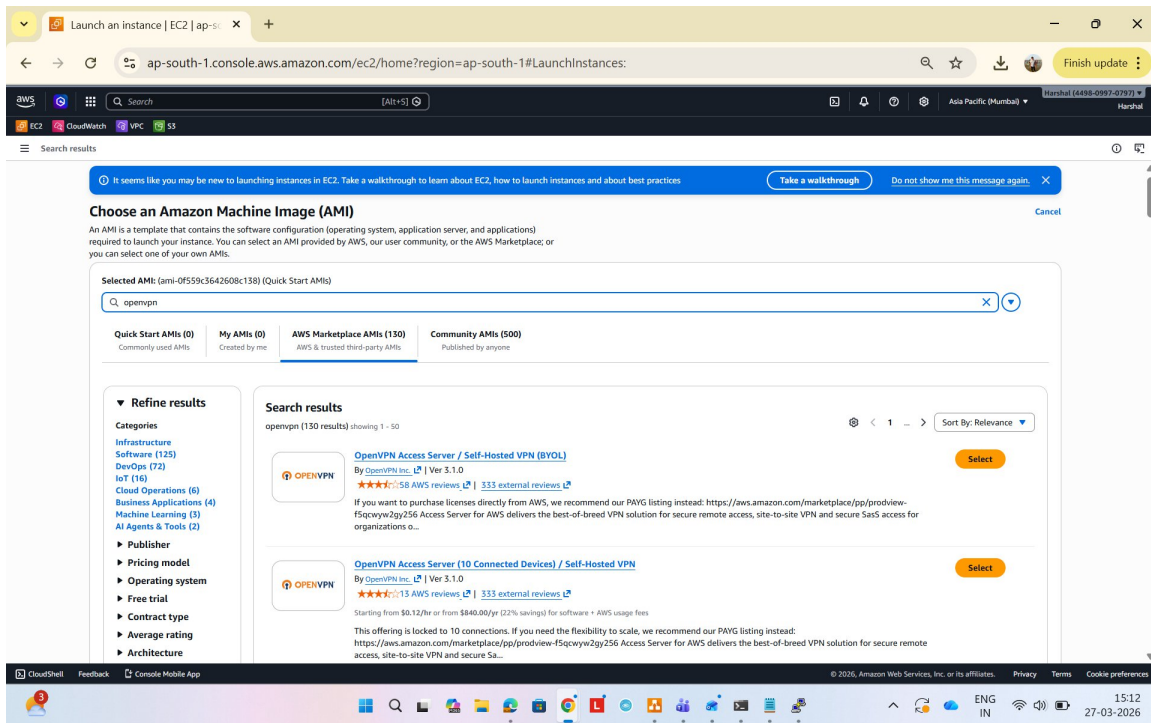
EC2 => Launch instance

The screenshot shows the 'Launch an instance' page in the AWS Management Console. The browser address bar displays 'ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:'. The page has a blue header with a 'Take a walkthrough' button. The main content area is titled 'Launch an instance' and includes a 'Name and tags' section with a text input field containing 'e.g. My Web Server'. Below this is the 'Application and OS Images (Amazon Machine Image)' section, which features a search bar and a grid of OS icons (Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, Debian). The 'Amazon Linux 2023 kernel-6.1 AMI' is selected. A 'Summary' panel on the right shows 'Number of instances: 1', 'Software Image (AMI): Amazon Linux 2023 AMI 2023.10', 'Virtual server type (instance type): t3.micro', 'Firewall (security group): New security group', and 'Storage (volumes): 1 volume(s) - 8 GiB'. At the bottom of the summary panel are 'Cancel' and 'Launch instance' buttons, along with a 'Preview code' link.

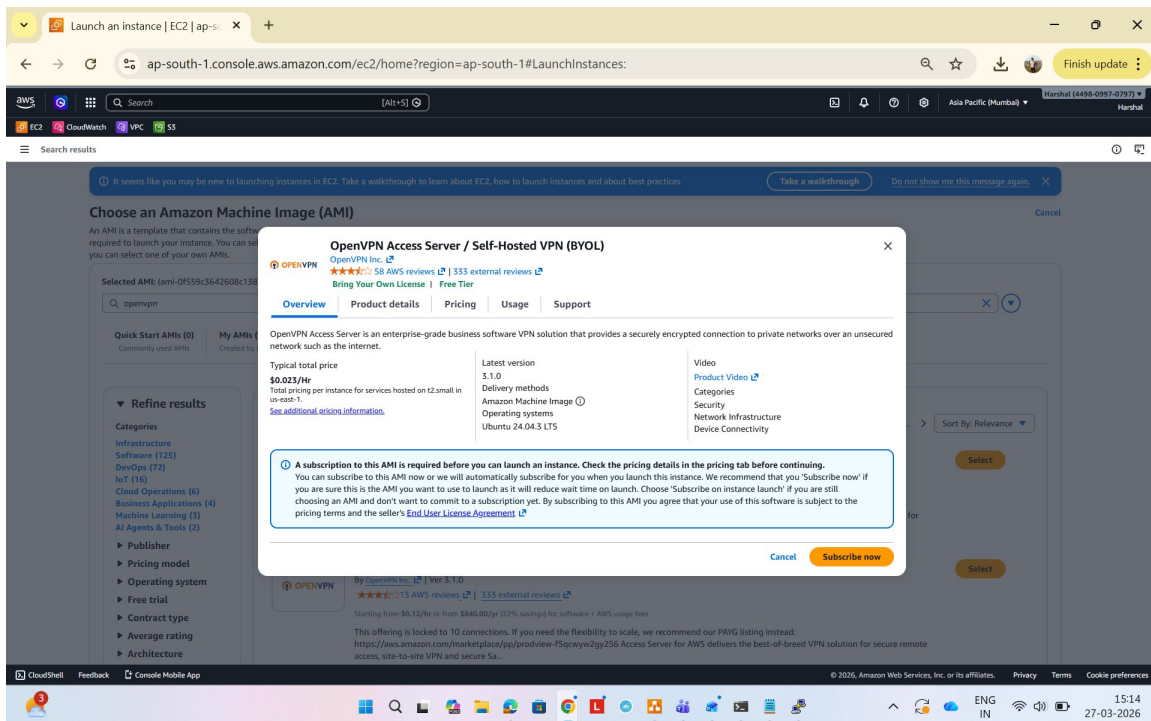
Enter Name: OpenVPN and click on Browse more AMIs

The screenshot shows the 'Choose an Amazon Machine Image (AMI)' page in the AWS Management Console. The browser address bar displays 'ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:'. The page has a blue header with a 'Take a walkthrough' button. The main content area is titled 'Choose an Amazon Machine Image (AMI)' and includes a search bar. Below the search bar are four tabs: 'Quick Start AMIs (44)', 'My AMIs (0)', 'AWS Marketplace AMIs (5973)', and 'Community AMIs (500)'. The 'Quick Start AMIs' tab is selected. The 'Refine results' section on the left shows filters for 'OS category' (All Linux/Unix, All Windows) and 'Architecture' (64-bit (Arm), 32-bit (x86), 64-bit (x86), 64-bit (Mac), 64-bit (Mac-Arm)). The 'All products (44 filtered, 44 unfiltered)' section displays a list of AMIs. The first two are 'Amazon Linux 2023 kernel-6.1 AMI' and 'Amazon Linux 2023 kernel-6.18 AMI'. The third is 'macOS Sequoia'. Each entry includes a 'Select' button. The 'macOS Sequoia' entry is highlighted.

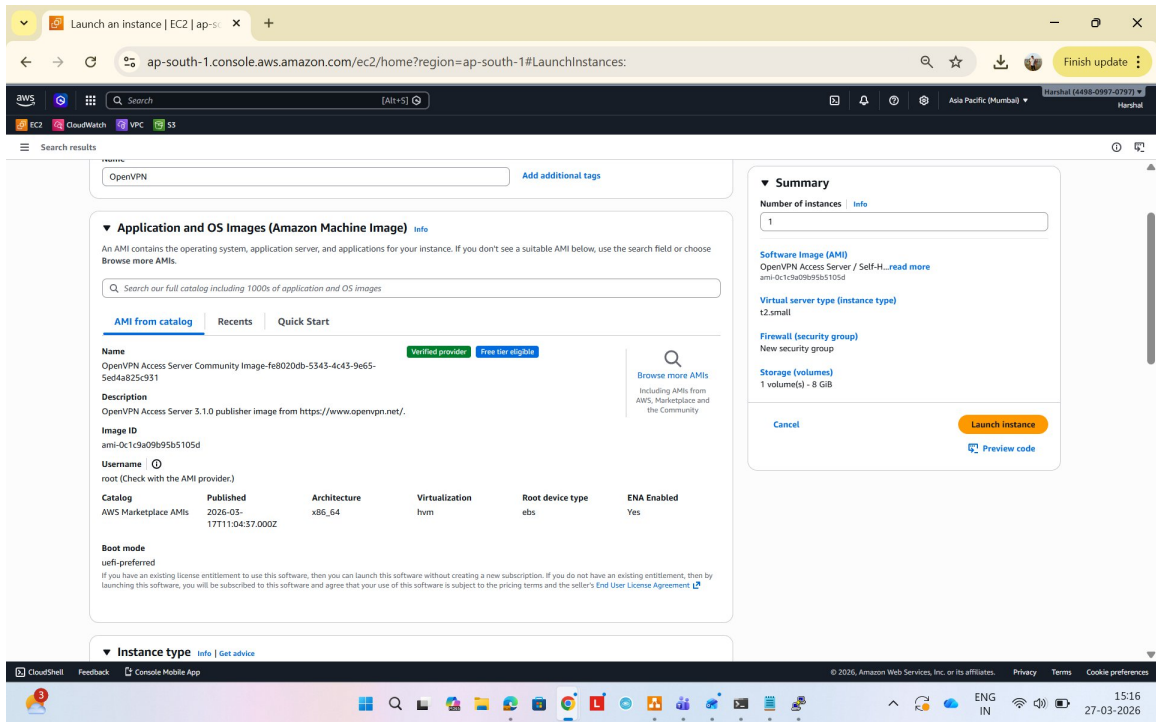
Click on AWS Marketplace AMIs and search openvpn



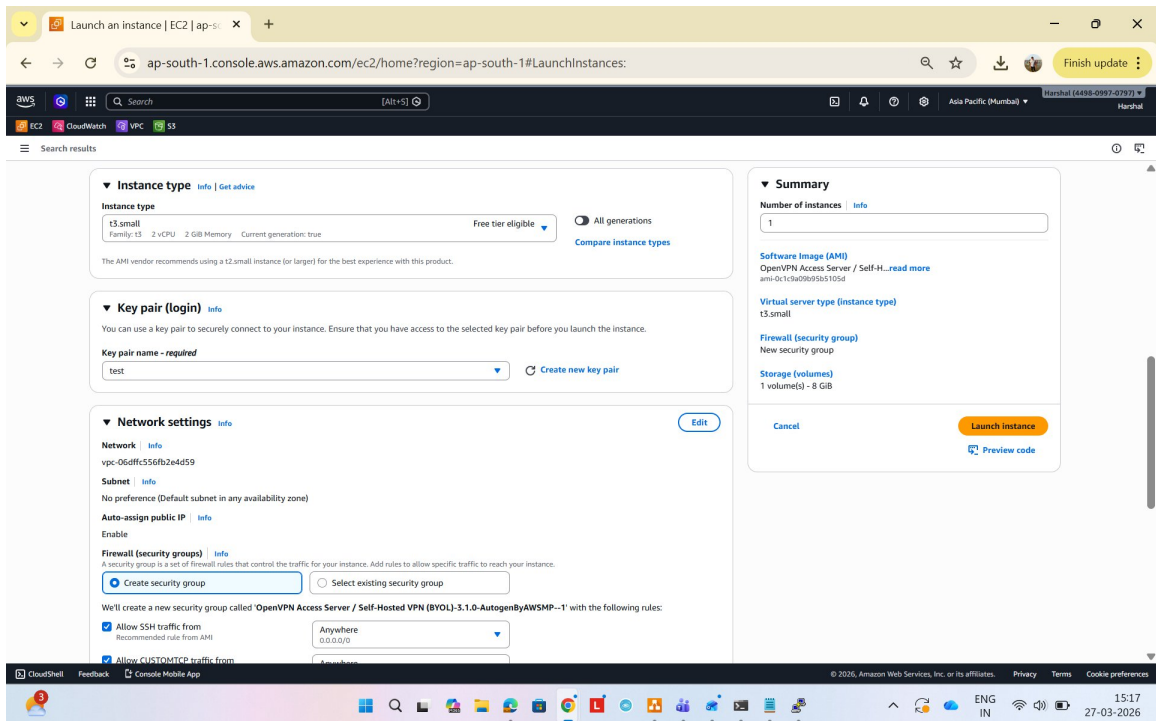
Click on first option



Click on Subscribe now.



Select instance type t3.small and click on create new key pair



In Network settings click on edit and select VPC

Subnet should be in public

Auto-assign public IP

Enable

Launch an instance | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Key pair name - required: test

Network settings

VPC - required: vpc-0746fca52911d55a9 (test)

Subnet: subnet-05de9fca4f98b95dd0

Auto-assign public IP: Enable

Firewall (security group): Create security group

Security group name - required: OpenVPN Access Server / Self-Hosted VPN (BYOL)-3.1.0-AutogenByAWSMP--2

Description - required: OpenVPN Access Server / Self-Hosted VPN (BYOL)-3.1.0-AutogenByAWSMP--2 created 2026-03-27T09

Inbound Security Group Rules

Security group rule 1 (TCP: 22, 0.0.0.0/0)

Type: ssh, Protocol: TCP, Port range: 22

Summary

Number of instances: 1

Software image (AMI): OpenVPN Access Server / Self-Hosted VPN

Virtual server type (instance type): t3.small

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 8 GiB

Launch instance

click on Launch Instance

Success

Successfully initiated launch of instance (i-0d5806aa236c5de5d)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
Test	i-0e47309eb410c9f4d	Stopped	t3.micro	3/3 checks passed	View alarms	ap-south-1a	-
OpenVPN	i-0d5806aa236c5de5d	Running	t3.small	3/3 checks passed	View alarms	ap-south-1c	-

Connect OpenVPN

Launch an Instance | EC2 | ap-south-1 | Connect to instance | EC2 | ap-south-1 | Instances | EC2 | ap-south-1 | +

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#ConnectToInstance:instanceId=i-05c4edb0ac142e032

Connect Info

Connect to an instance using the browser-based client.

Instance ID i-05c4edb0ac142e032 (OpenVPN)	VPC ID vpc-0746fca52911d53a9	Security groups sg-0320ba250b84dbb20 (OpenVPN Access Server / Self-Hosted VPN (BYOL)-3.1.0-AutoGenByAWSMP-2)	IAM role -
--	---------------------------------	---	---------------

EC2 Instance Connect | SSM Session Manager | SSH client | EC2 serial console

Instance ID
i-05c4edb0ac142e032 (OpenVPN)

Connection type

☒ Connect using a Public IP
Connect using a public IPv4 or IPv6 address

☐ Connect using a Private IP
Connect using a private IP address and a VPC endpoint

☒ Public IPv4 address
65.1.64.206

☐ IPv6 address

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, root.

root

Note: In most cases, the default username, root, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel Connect



Launch an Instance | EC2 | ap-south-1 | Instances | EC2 | ap-south-1 | EC2 Instance Connect | ap-south-1 | Instances | EC2 | ap-south-1 | +

ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipV4&connType=standard&instanceId=i-05c4ed...

EC2 Instance Connect

Public IP: 65.1.64.206 Private IP: 10.0.9.223

Please enter 'yes' to indicate your agreement [no]:

i-05c4edb0ac142e032 (OpenVPN)

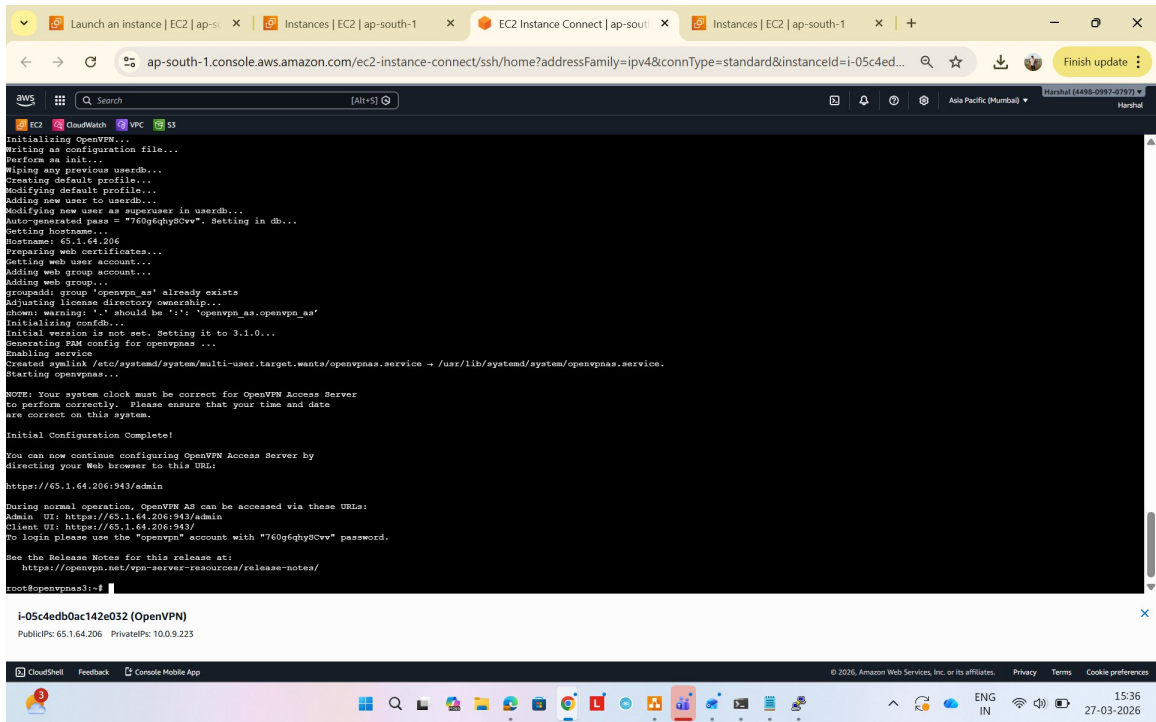
PublicIPs: 65.1.64.206 PrivateIPs: 10.0.9.223

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

ENG IN 15:34 27-03-2026

Enter yes and enter for all the remaining instructions



```
Initializing OpenVPN...
Writing configuration file...
Performing init...
Wiping any previous users...
Creating default profile...
Modifying default profile...
Adding new user to userdb...
Modifying new user as superuser in userdb...
Auto-generated pass = "760g6qhySCvv". Setting in db...
Getting hostname...
Hostname: 65.1.64.206
Preparing web certificates...
Setting web user account...
Adding web group account...
Adding web group...
groupadd: group 'openvpn_as' already exists
Adjusting license directory ownership...
chown: warning: '.' should be '.'; 'openvpn_as.openvpn_as'
Initializing confd...
Initial version is not set. Setting it to 3.1.0...
Generating FPM config for openvpnas ...
Enabling service
Created symlink /etc/systemd/system/multi-user.target.wants/openvpnas.service → /usr/lib/systemd/system/openvpnas.service.
Starting openvpnas...

NOTE: Your system clock must be correct for OpenVPN Access Server
to perform correctly. Please ensure that your time and date
are correct on this system.

Initial Configuration Complete!

You can now continue configuring OpenVPN Access Server by
directing your Web browser to this URL:

https://65.1.64.206:943/admin

During normal operation, OpenVPN AS can be accessed via these URLs:
Admin UI: https://65.1.64.206:943/admin
Client UI: https://65.1.64.206:943/
To login please use the "openvpn" account with "760g6qhySCvv" password.

See the Release Notes for this release at:
https://openvpn.net/vpn-server-resources/release-notes/

root@openvpnas3:~#
```

i-05c4edb0ac142e032 (OpenVPN)
PublicIPs: 65.1.64.206 PrivateIPs: 10.0.9.225

Copy last part from output

You can now continue configuring OpenVPN Access Server by directing your Web browser to this URL:

`https://65.1.64.206:943/admin`

During normal operation, OpenVPN AS can be accessed via these URLs:

Admin UI: `https://65.1.64.206:943/admin`

Client UI: `https://65.1.64.206:943/`

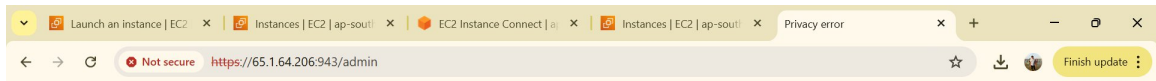
To login, please use the "openvpn" account with "760g6qhySCvv" password.

See the Release Notes for this release at:

`https://openvpn.net/vpn-server-resources/release-notes/`

`root@openvpnas3:~#`

Copy Admin UI url and browse



Your connection is not private

Attackers might be trying to steal your information from **65.164.206** (for example, passwords, messages, or credit cards). [Learn more about this warning](#)

NET:ERR_CERT_AUTHORITY_INVALID



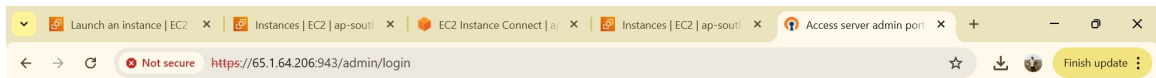
[Turn on enhanced protection](#) to get Chrome's highest level of security

Advanced

Back to safety



Click on advance and proceed



Admin Login

Username *

Password *

Sign In



Enter Username and password which we got from configuration

The image shows two screenshots of the OpenVPN Access Server interface. The top screenshot is the 'Admin Login' page, which has a light blue background with a grid pattern. It features the 'Access Server' logo at the top. The login form is centered and contains a 'Username' field with the value 'openvpn', a 'Password' field with masked characters, and a 'Sign In' button. The browser's address bar shows the URL 'https://65.164.206.943/admin/login'. The bottom screenshot shows the 'License Agreement' dialog box over a blurred background of the server's status page. The dialog is titled 'License Agreement' and contains the 'OpenVPN Access Server End User License Agreement (OpenVPN-AS EULA)'. It lists four terms: 1. Copyright Notice, 2. Redistribution, 3. Agreement not to reverse engineer, and 4. Bundling with open source software. Below the terms, there is a checkbox area with 'I have read and agree to the terms of the OpenVPN Access Server End User License Agreement above.' and two buttons: 'Agree' and 'I Do Not Accept'. The browser's address bar in this screenshot shows 'https://65.164.206.943/admin/status'.

Access Server

Admin Login

Username *

openvpn

Password *

.....

Sign In

License Agreement

OpenVPN Access Server End User License Agreement (OpenVPN-AS EULA)

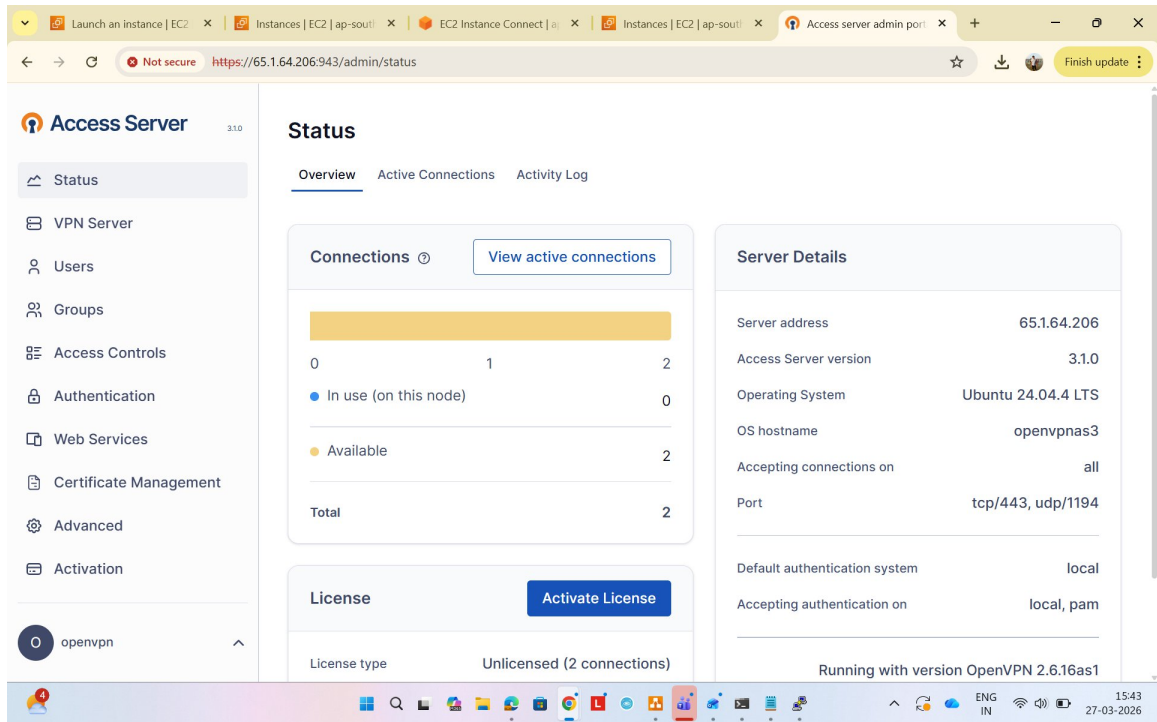
1. Copyright Notice: OpenVPN Access Server License;
Copyright (c) 2009-2026 OpenVPN Inc. All rights reserved.
"OpenVPN" is a trademark of OpenVPN Inc.
2. Redistribution of OpenVPN Access Server binary forms and related documents,
are permitted provided that redistributions of OpenVPN Access Server binary
forms and related documents reproduce the above copyright notice as well as
a complete copy of this EULA.
3. You agree not to reverse engineer, decompile, disassemble, modify,
translate, make any attempt to discover the source code of this software,
or create derivative works from this software.
4. The OpenVPN Access Server is bundled with other open source software

I have read and agree to the terms of the OpenVPN Access Server End User License Agreement above.

Agree

I Do Not Accept

Click on Agree



Done.

Download and install OpenVPN on your system

Link: <https://openvpn.net/downloads/openvpn-connect-v3-windows.msi>

OpenVPN Connect

←

?

Connect with **Server URL** or **Cloud ID** provided by your Admin

🌐

Enter your URL or Cloud ID

Continue →

Where to get my URL?

Have a configuration file instead (.ovpn)?

📁

Upload File

Enter URL: `https://65.1.64.206:943/admin`
and click Continue.

OpenVPN Connect

←

?

Username *

Username

Password *

Password

Profile Details

Profile Name *

13.235.113.105

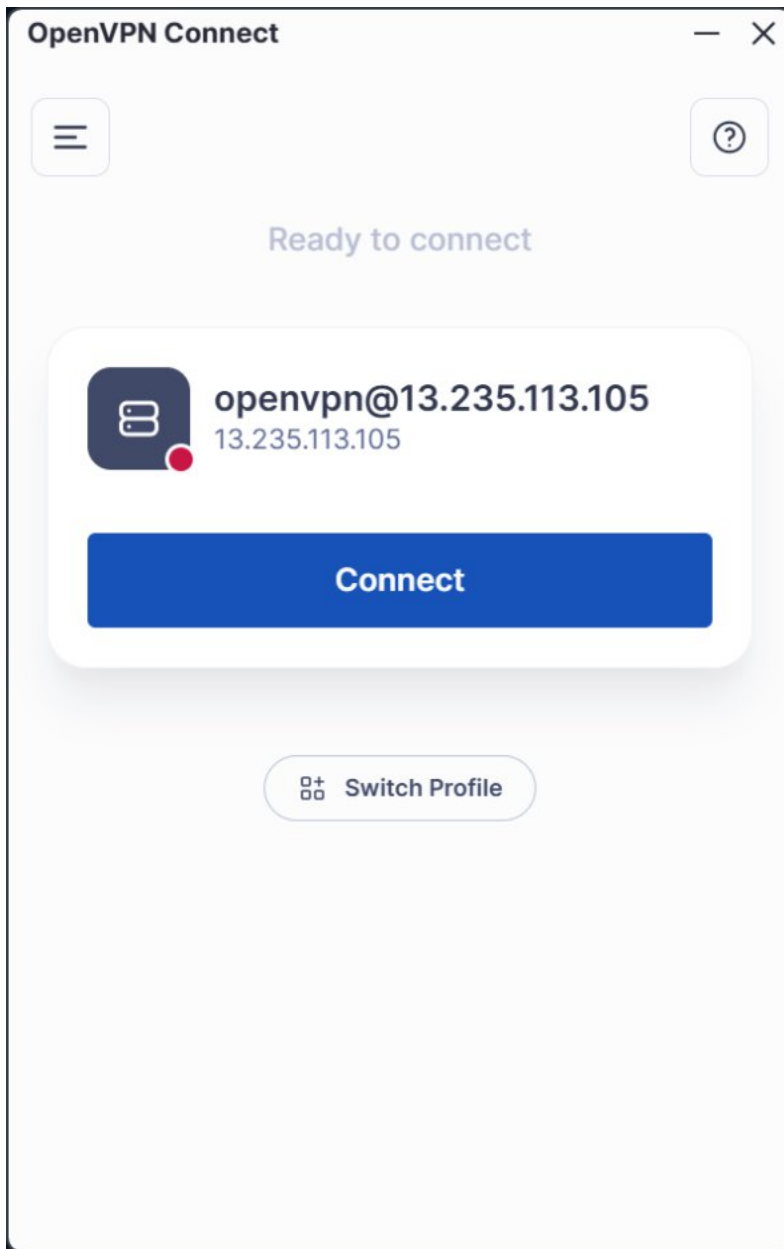
Hostname

13.235.113.105

Import

Enter username: openvpn
password: 760g6qhySCvv

Click on Import



Click on connect

Enter password



Profile:

openvpn@13.235.113.105

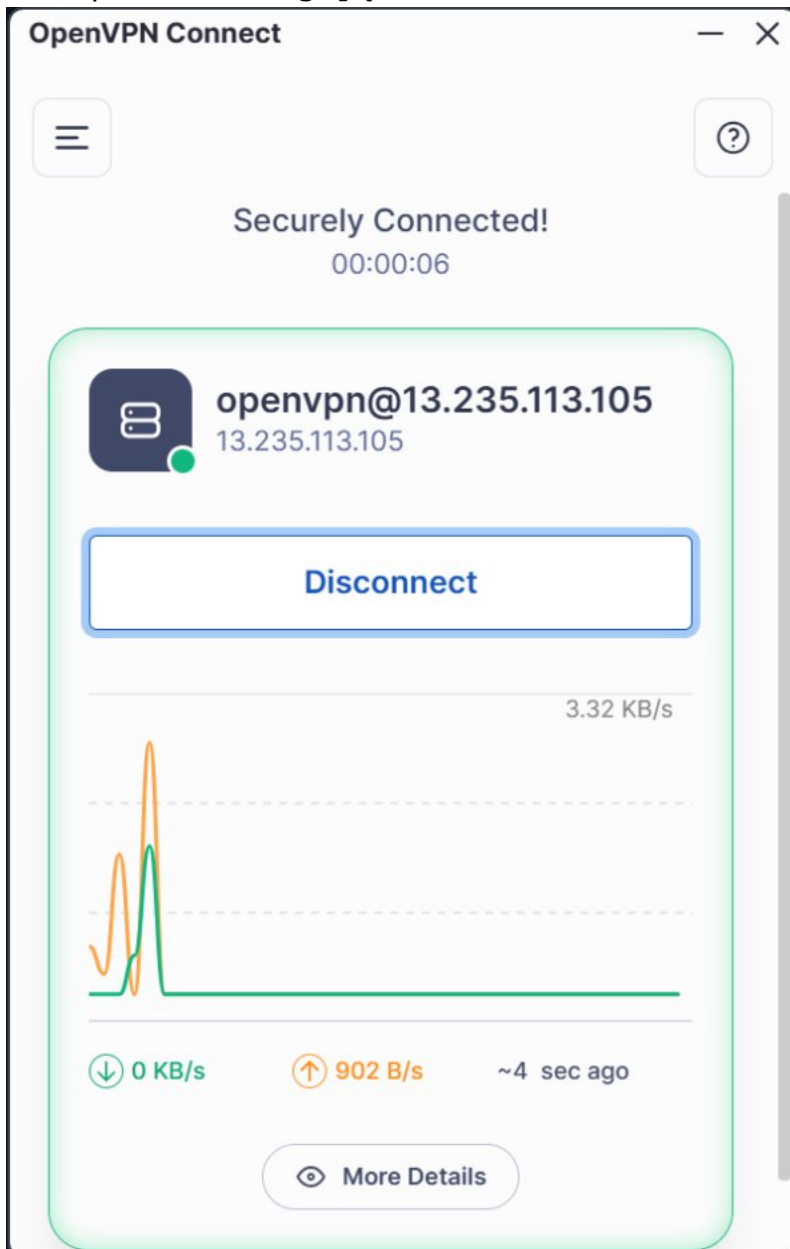
Password



Enter

Cancel

Enter password: 760g6qhySCvv



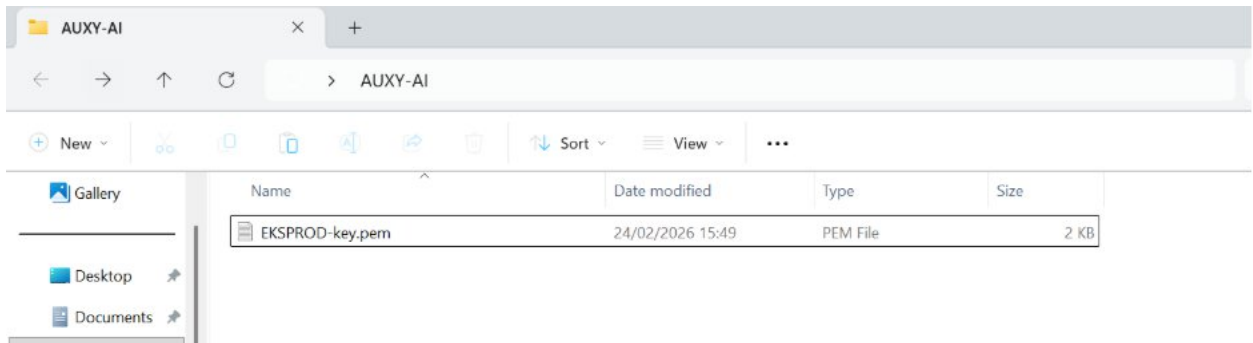
OpenVPN is connected.

Open Command prompt

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\kshitija.chavan>
```

Go to the pem file location



And run below command (Private instance)

```
ssh -i "C:\Users\harshal.wakde\Downloads\EKSPROD-key.pem"
ubuntu@10.0.136.242
```

```
ubuntu@ip-10-0-136-242: ~  
C:\Users\kshitija.chavan>ssh -i "C:\Users\kshitija.chavan\Downloads\EKSPROD-key.pem" ubuntu@10.0.136.242  
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.17.0-1007-aws x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Tue Feb 24 13:10:29 UTC 2026  
  
System load:  0.0      Temperature:    -273.1 C  
Usage of /:   28.7% of 47.39GB  Processes:      126  
Memory usage: 17%      Users logged in: 1  
Swap usage:   0%        IPv4 address for ens5: 10.0.136.242  
  
* Ubuntu Pro delivers the most comprehensive open source security and compliance features.  
  
https://ubuntu.com/aws/pro  
  
Expanded Security Maintenance for Applications is not enabled.  
  
52 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
1 additional security update can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
Last login: Tue Feb 24 12:42:17 2026 from 10.0.13.122  
ubuntu@ip-10-0-136-242:~$
```

=====

Create IAM User

Requirement:

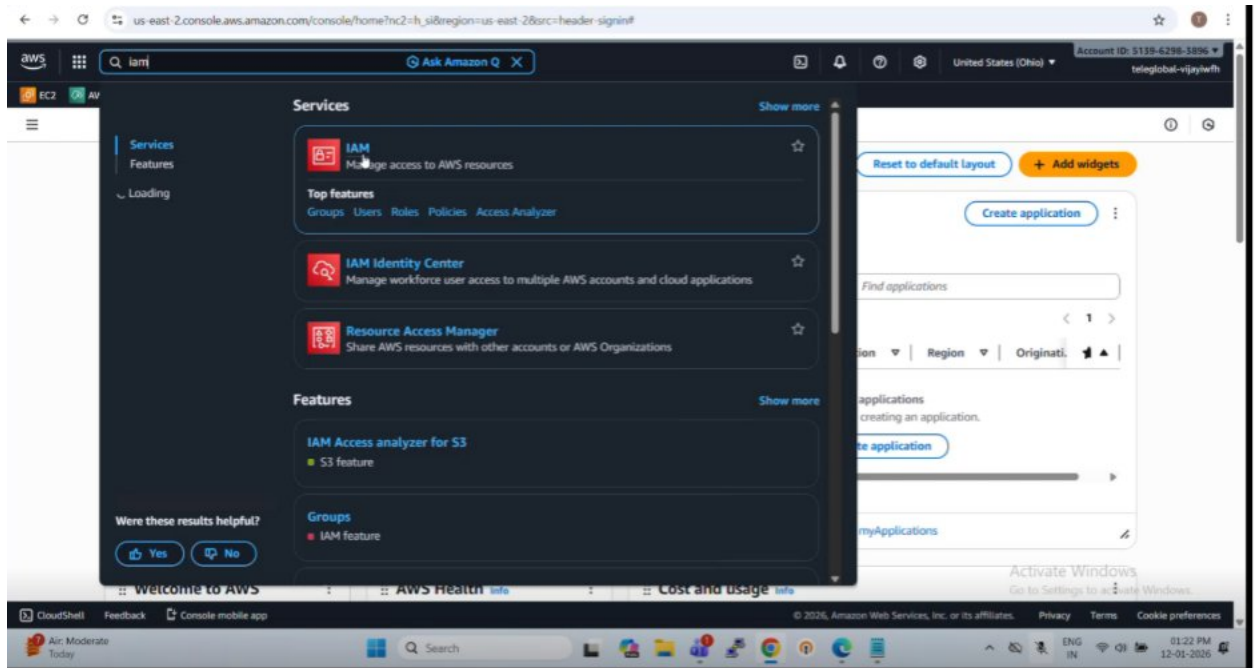
2. S3 Buckets and IAM User Setup

Separate S3 buckets will be maintained for **Development** and **Production** environments. Corresponding IAM users will be created with **bucket-specific access** as outlined below:

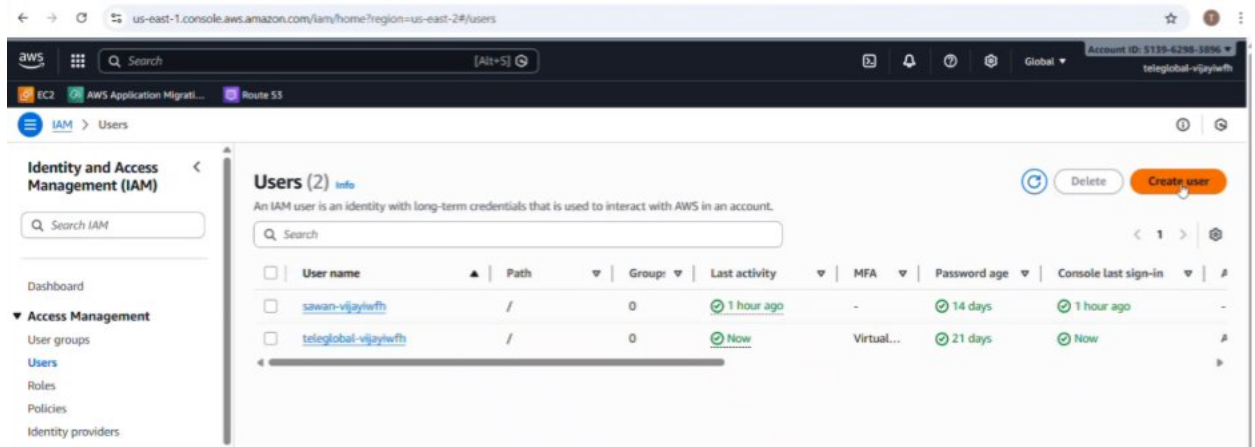
- **VFHDEVS3APP01**
IAM User: `tse_dev_s3_service_integrator`
Permissions: PutObject and GetObject access will be granted only to the Dev S3 bucket.
- **VFHPRODS3APP01**
IAM User: `tse_prod_s3_service_integrator`
Permissions: PutObject and GetObject access will be granted only to the Prod S3 bucket..

Appropriate permissions will be granted to each IAM user based on their respective bucket access requirements.

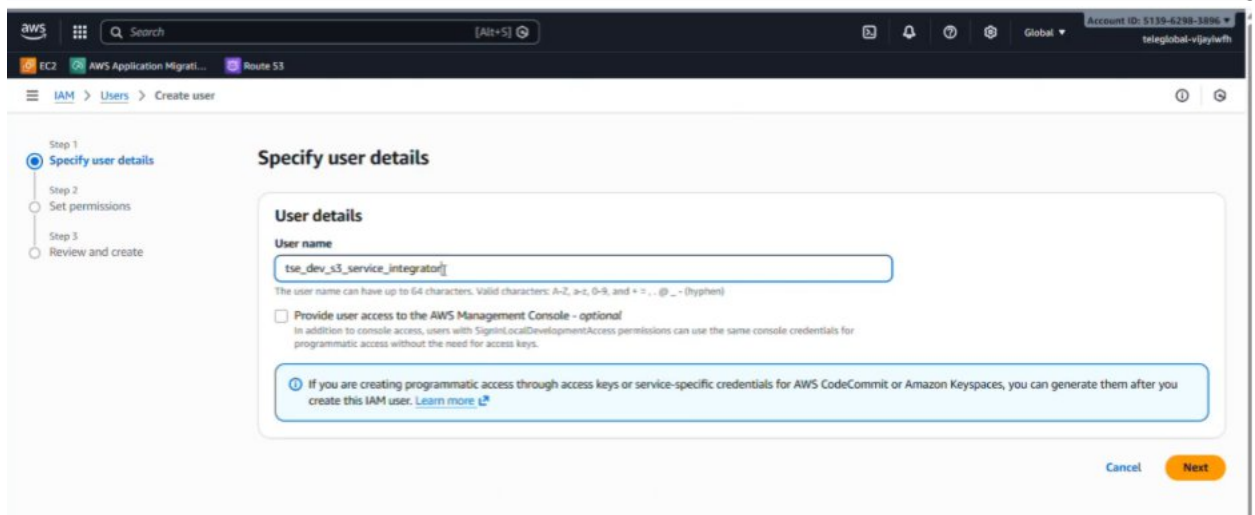
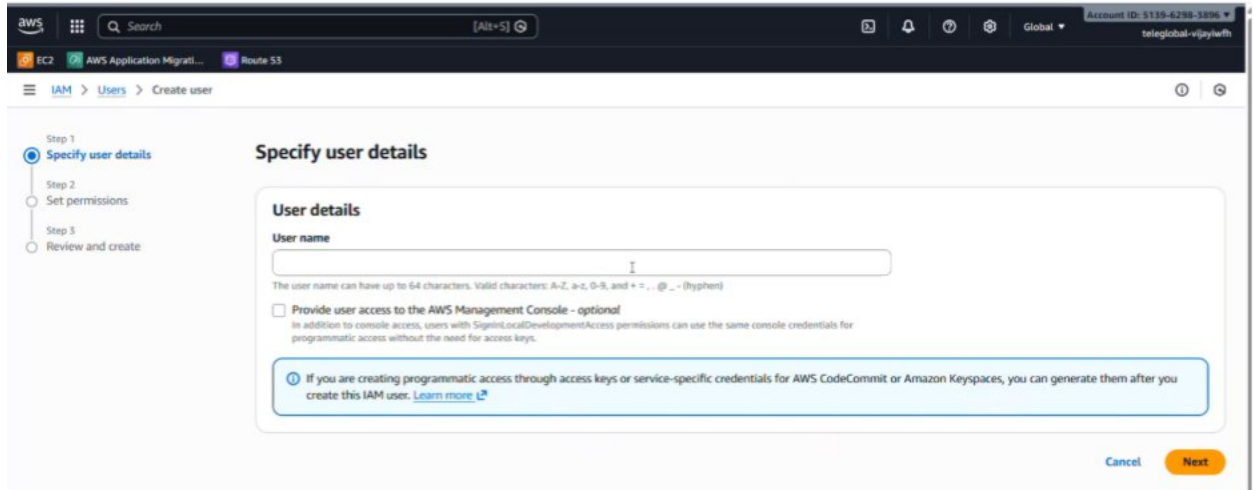
1. Log in to the AWS Management Console.
2. In the search bar, type **IAM** and open the IAM service.



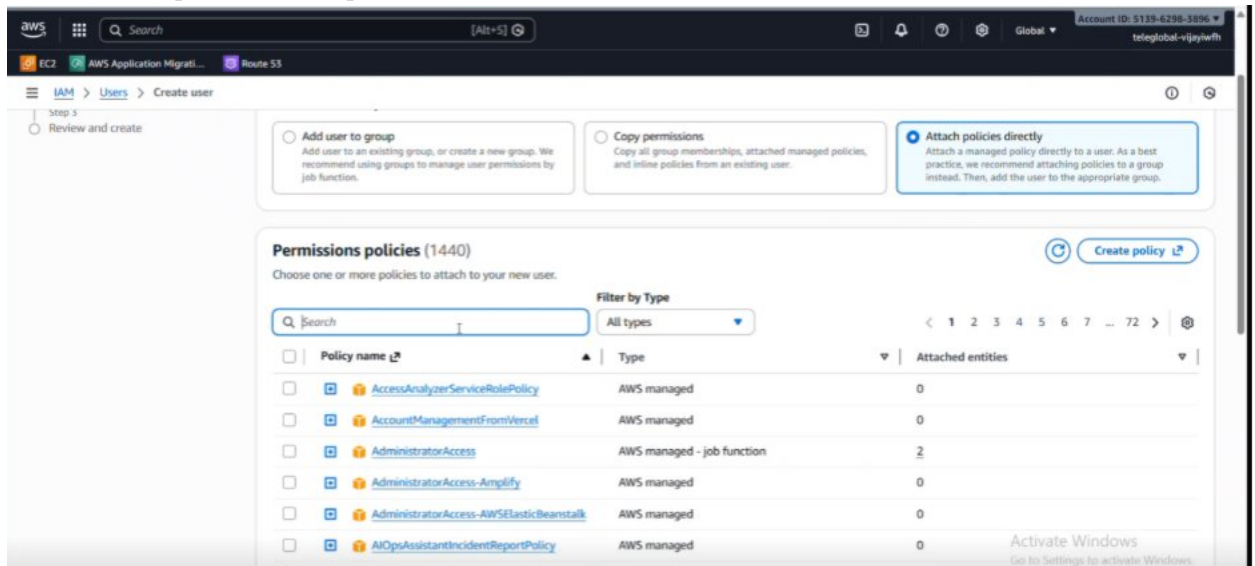
3. In the left navigation pane, select **Users**.
4. Click **Create user**.



5. Enter the **User name** as per naming standards.



6. Click **Next** to proceed with permissions.



7. Select **Attach policies directly**.

Create and Attach Custom IAM Policy

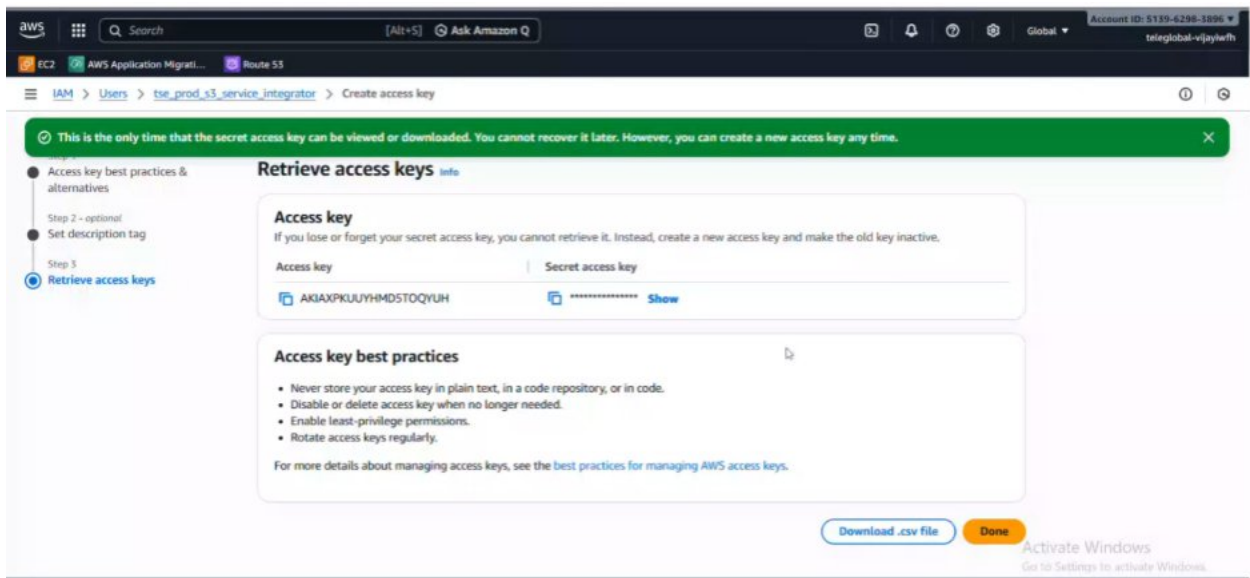
1. Click **Create policy**.
2. Select the **JSON** tab.
3. Insert the following policy document. Replace the bucket name if required:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowPutGetObjectsOnly",
      "Effect": "Allow",
      "Action": [
        "s3:PutObject",
        "s3:GetObject"
      ],
      "Resource": "arn:aws:s3:::vfhdevs3app01/*"
    }
  ]
}
```

4. Click **Next**.
5. Enter the **Policy name**: AllowPutGetObjectsOnly.
6. Review the policy and click **Create policy**.
7. Return to the IAM user creation screen.
8. Search for and select the policy **AllowPutGetObjectsOnly**.
9. Click **Next** and then **Create the user**.

Generate Access Credentials

1. Open the newly created IAM user.
2. Navigate to the **Security credentials** tab.
3. Under **Access keys**, click **Create access key**.
4. Select the appropriate use case and proceed.
5. Securely download and store:
 - a. Access Key ID
 - b. Secret Access Key



Note: Access keys must be stored securely and must not be shared over email or unsecured channels.

S3 Folder Structure Configuration

Requirement:

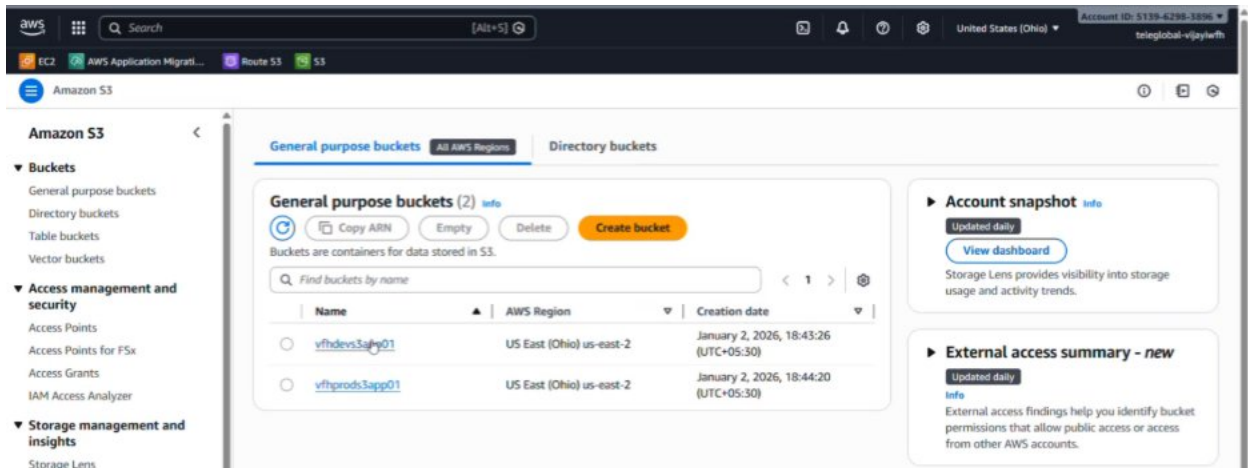
3. S3 Folder Structure

As discussed, the required folder structure will be created within the S3 buckets as per the shared screenshot and aligned with application requirements:

```
tse_attach
├── work_item
│   ├── attachment
│   └── thumbnail
├── comment
│   ├── attachment
│   └── thumbnail
├── leaves
│   ├── attachment
│   └── thumbnail
├── chat
│   ├── attachment
│   └── thumbnail
```

Access Target S3 Bucket

1. Open the **S3** service from the AWS Console.
2. Select the target bucket.



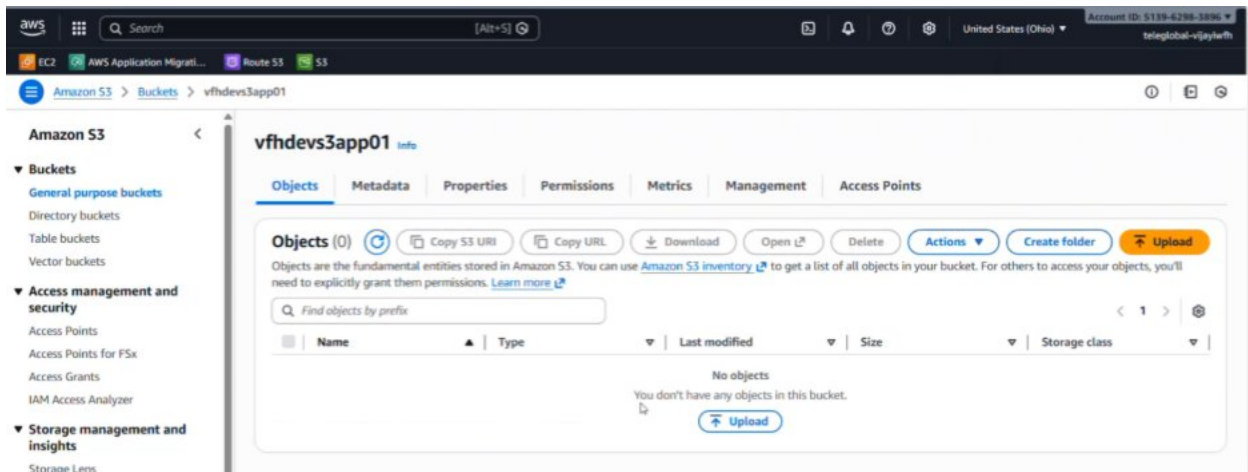
• S3 1.1

Create Root-Level Folders

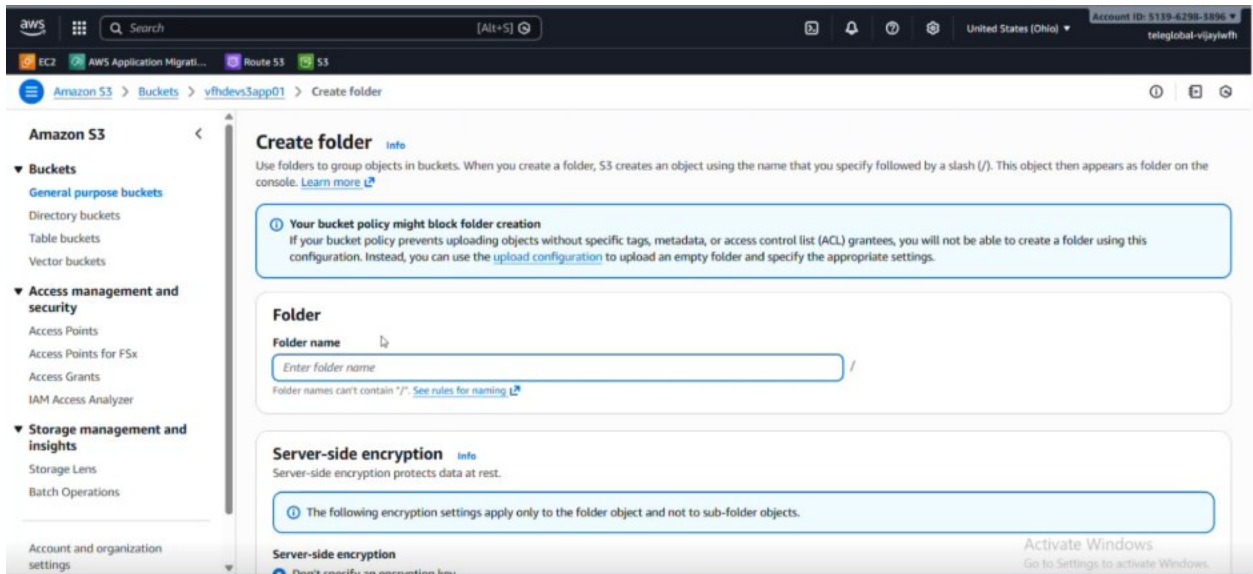
- tse_attach

For each root folder:

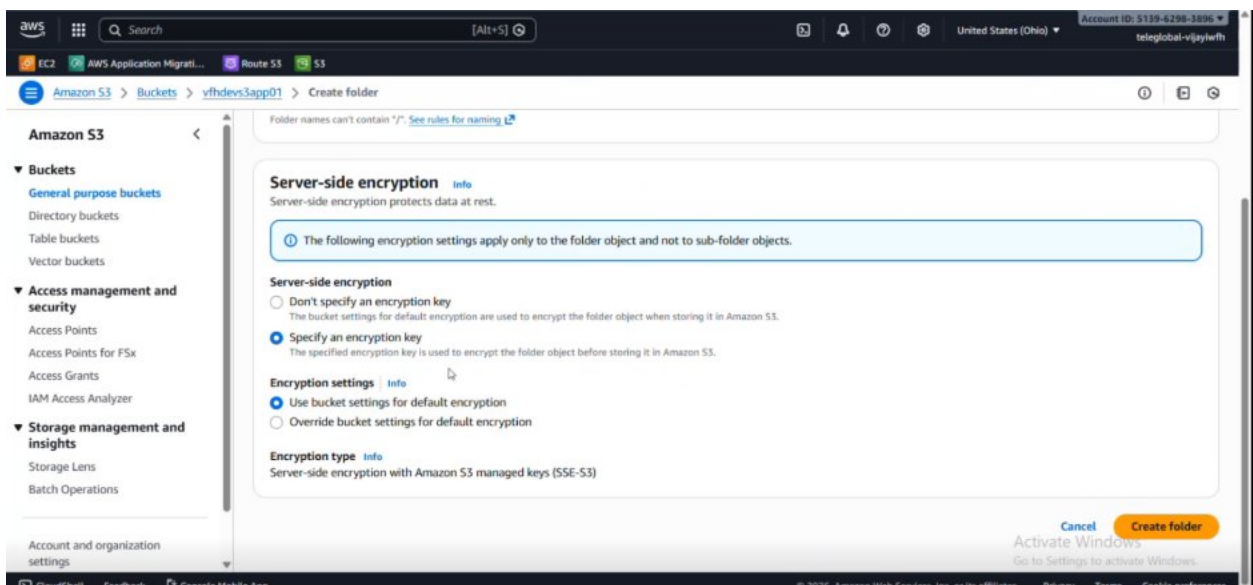
1. Click the **Create folder**.



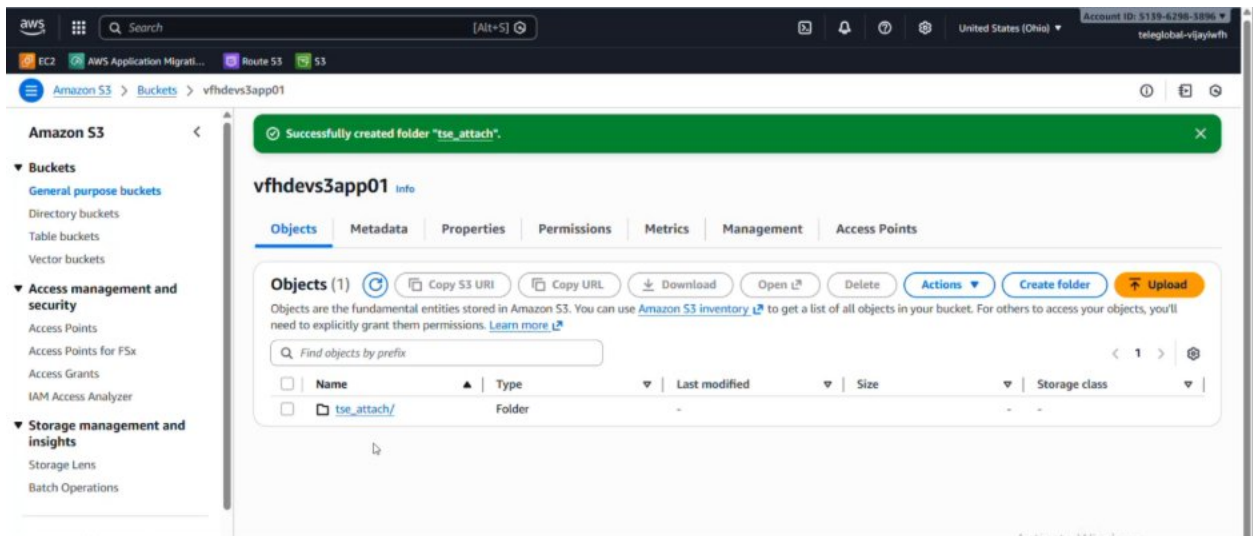
2. Enter the folder name.



3. Click on Specify the required encryption key.



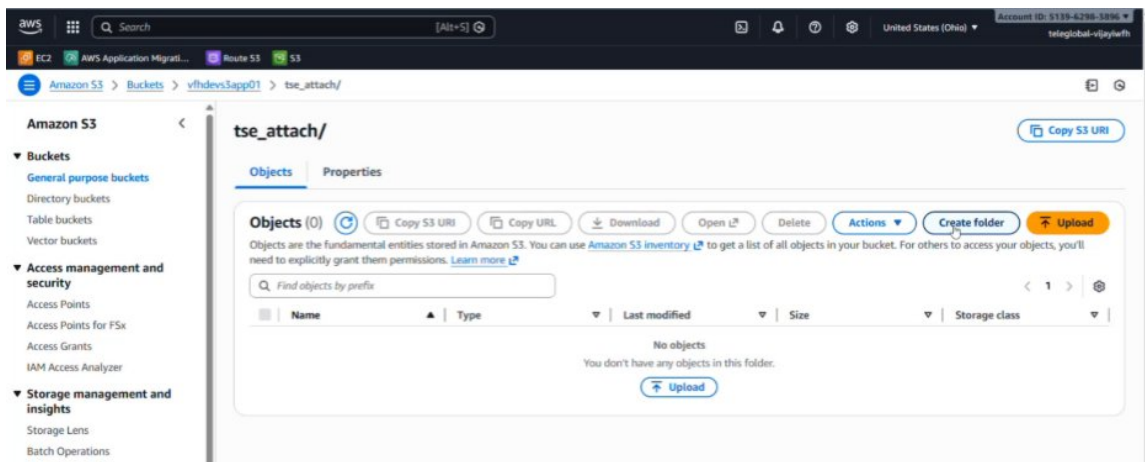
4. Click the **Create folder** to confirm.



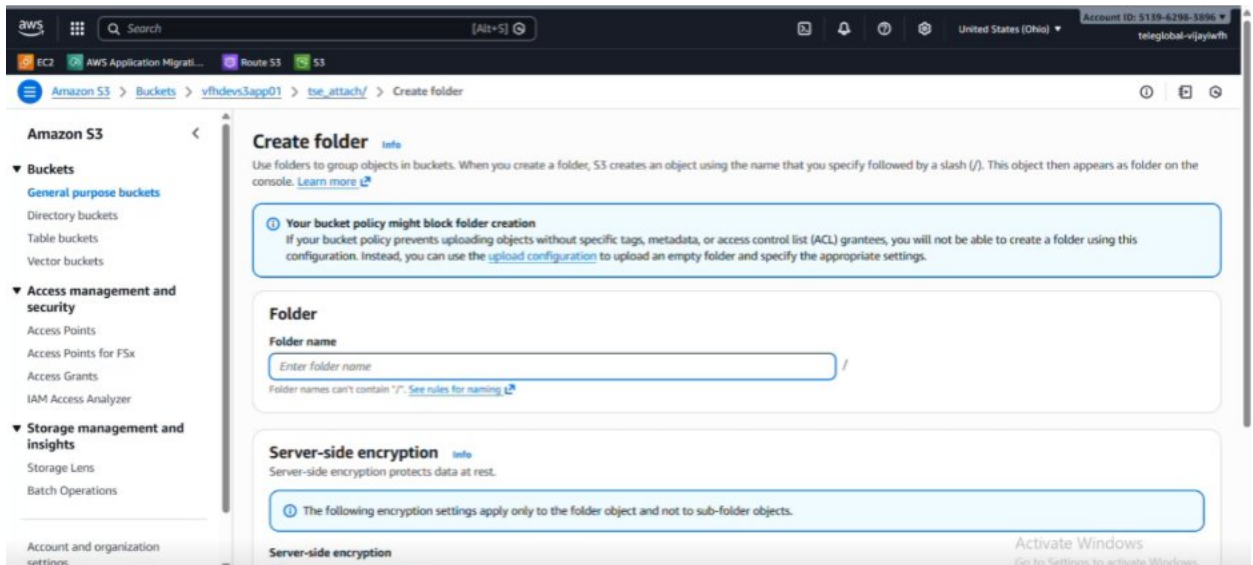
Create sub-folders

- Work_item
- comment
- leaves
- Chat

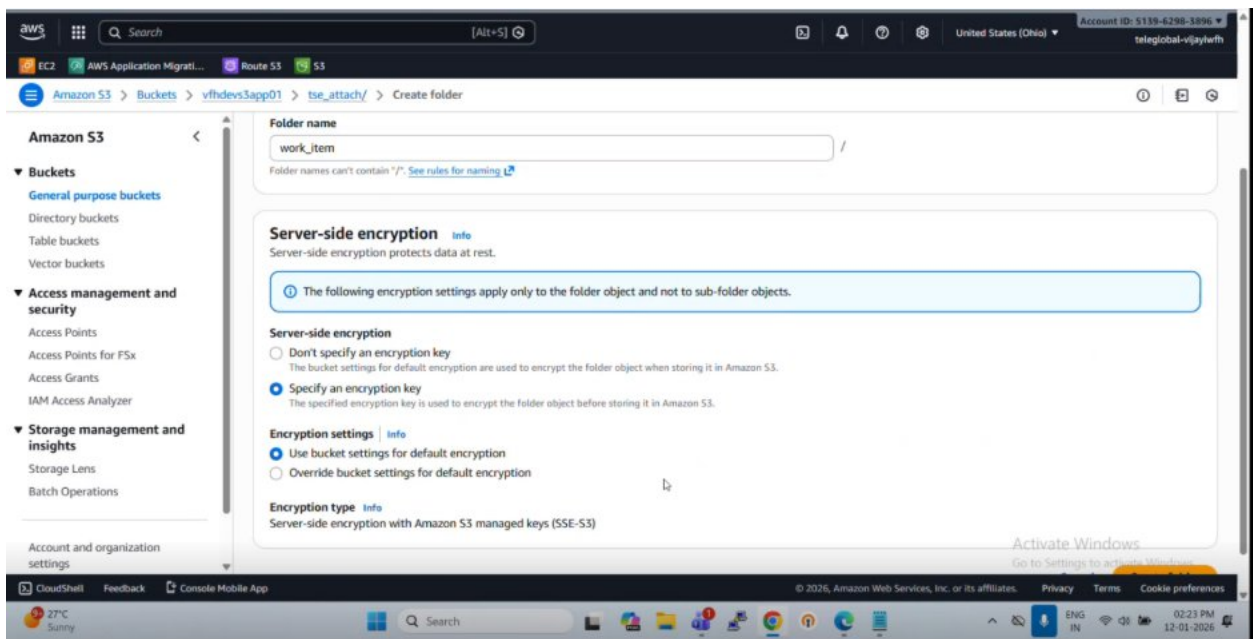
1. Open the required root folder.



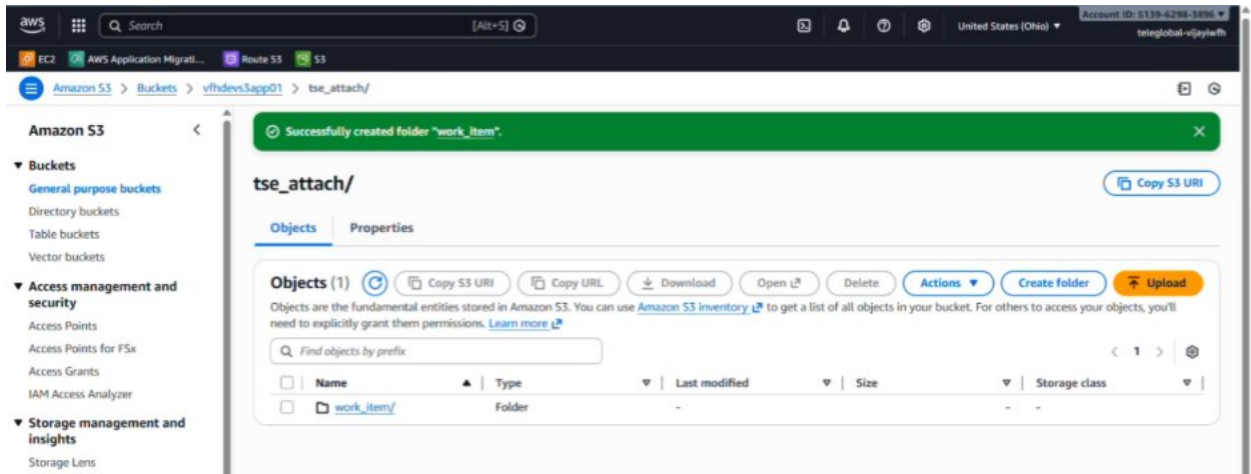
2. Click **Create folder**.



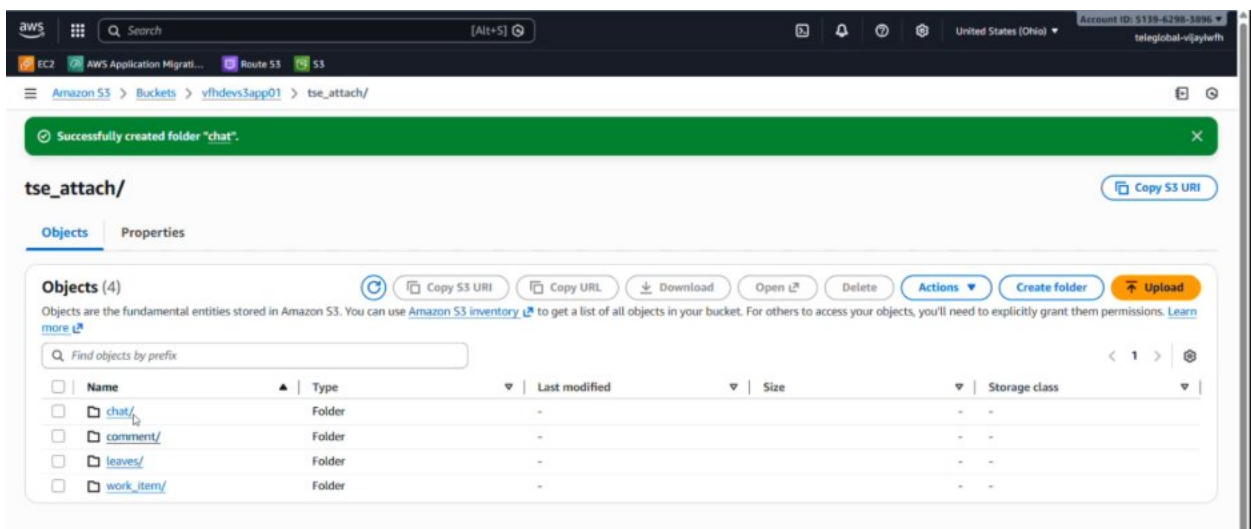
3. Enter the sub-folder name and click on **specify an encryption key**.



4. Click **Create folder**.



5. Repeat for all required sub-folders.



Create Sub-Folder

Within each root folder, create the approved sub-folders as listed below.

Standard Sub-Folders:

- attachment
- thumbnail

Steps:

6. Open the required root folder.

7. Click **Create folder**.
8. Enter the sub-folder name and click on **specify an encryption key**.
9. Click **Create folder**.
10. Repeat for all required sub-folders.

Replication to Second Bucket

1. Open the second S3 bucket.
2. Repeat all steps from **S3 1.1**.
3. Ensure the folder and sub-folder structure exactly match the primary bucket.

Procedure to Create Hosted Zone and Connect Domain to ALB

Purpose

This section defines the standard procedure to create a hosted zone in Amazon Route 53 after receiving the domain name from the client, share name server details with the client, and connect the hosted zone to the Application Load Balancer (ALB).

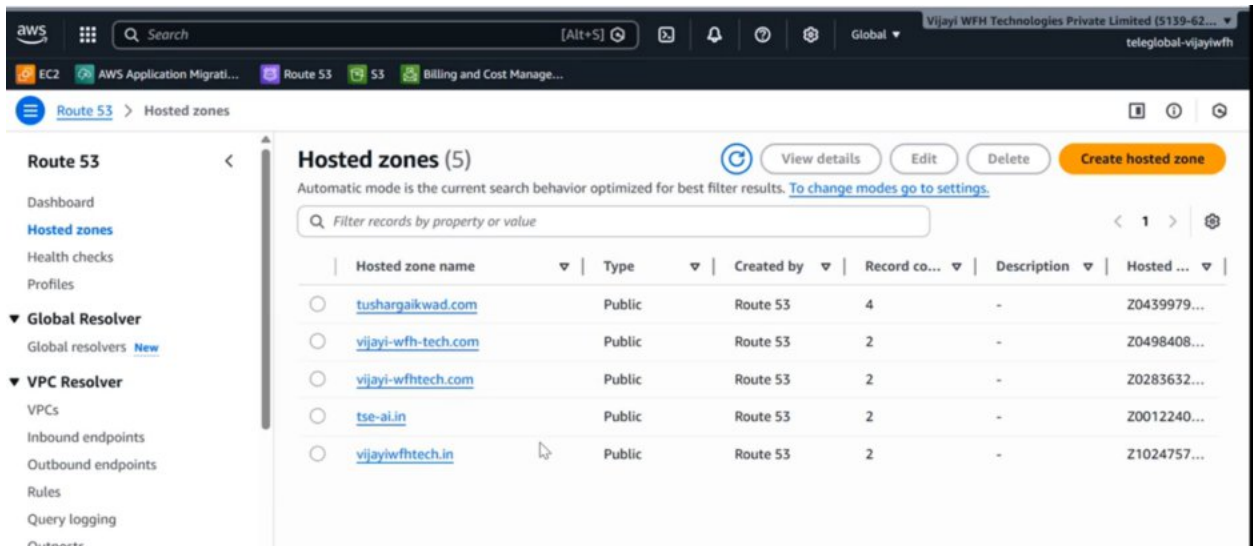
Prerequisite

- Domain name provided by the client.
- Access to AWS Console with Route 53 and ELB permissions.
- Access to client DNS registrar portal (or coordination with client team), e.g., GoDaddy.

Create Hosted Zone in Route 53

Access Route 53

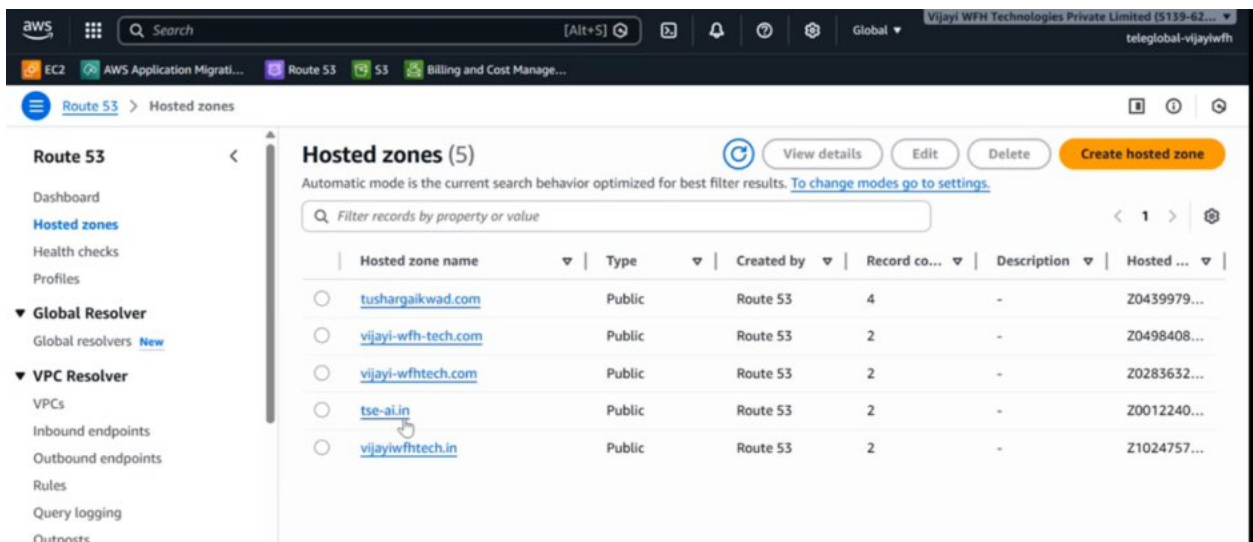
1. Log in to the AWS Management Console.
2. In the search bar, type **Route 53** and open the service.
3. In the left navigation pane, select **Hosted zones**.



4. Click **Create hosted zone**.

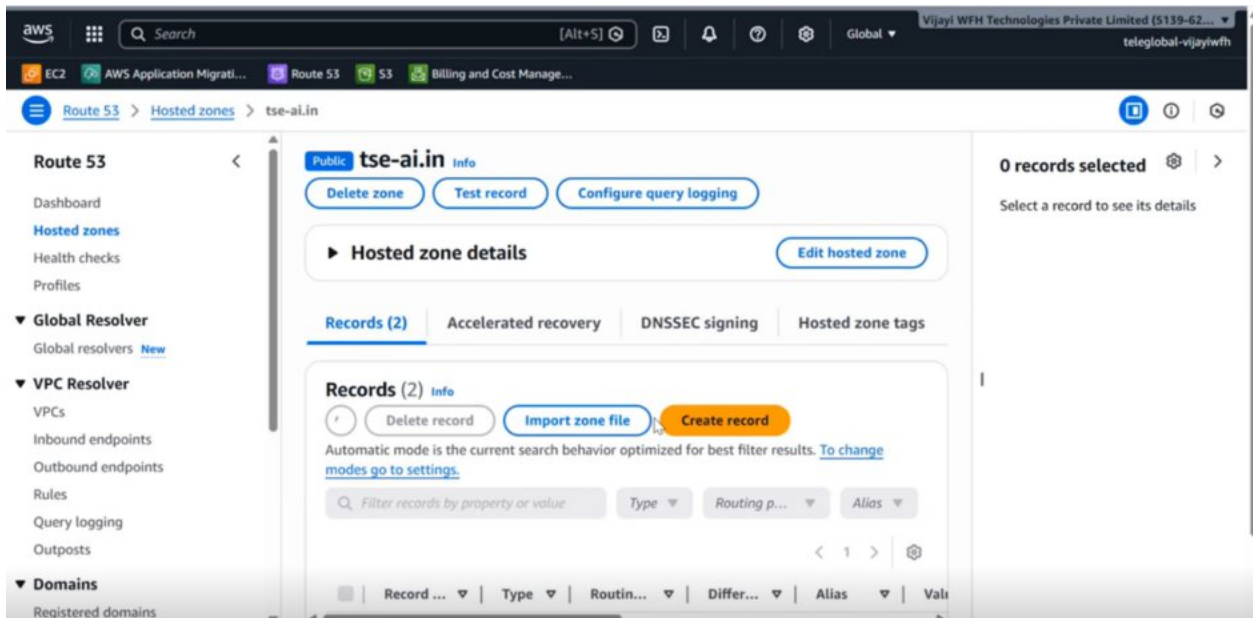
Enter Hosted Zone Details

1. Enter the **Domain name** as provided by the client.
2. Enter a brief **Description** (as per internal standards).
3. Select **Public hosted zone**.
4. Click **Create hosted zone** to complete the creation.

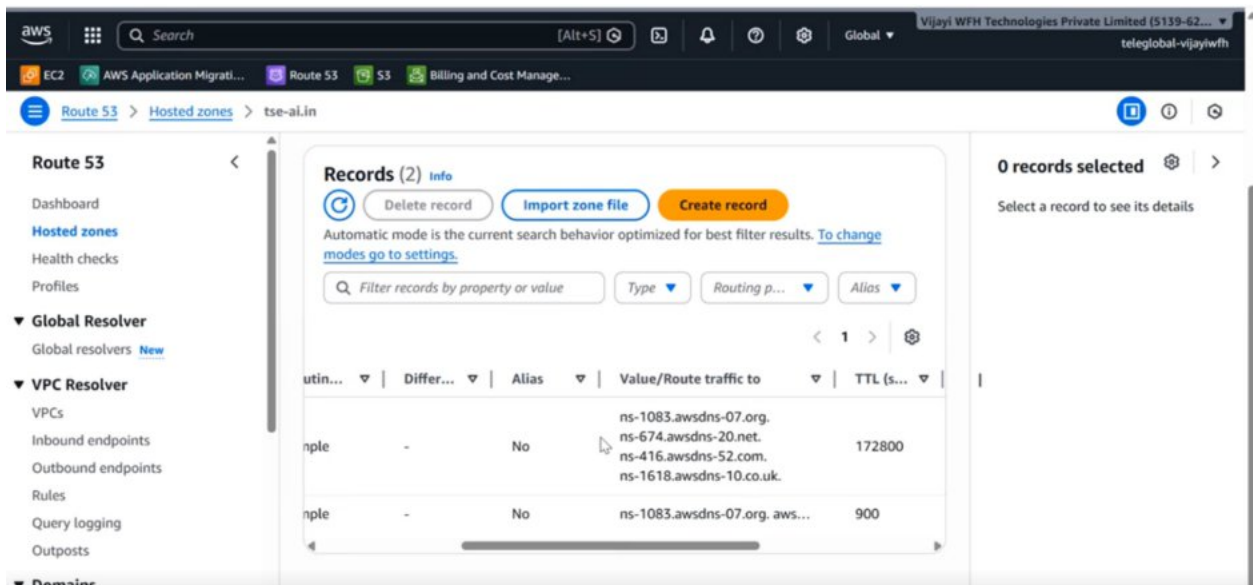


Share Name Server Details with Client

1. Open the newly created hosted zone.



2. Locate the **Name Server (NS)** records.



3. Copy all assigned name server values.

4. Share these name server values with the client via approved communication channels.

Client DNS Update (Registrar Level)

1. Client updates the domain name server entries at their registrar (e.g., GoDaddy).
2. Client replaces existing name servers with the AWS Route 53 name servers provided.

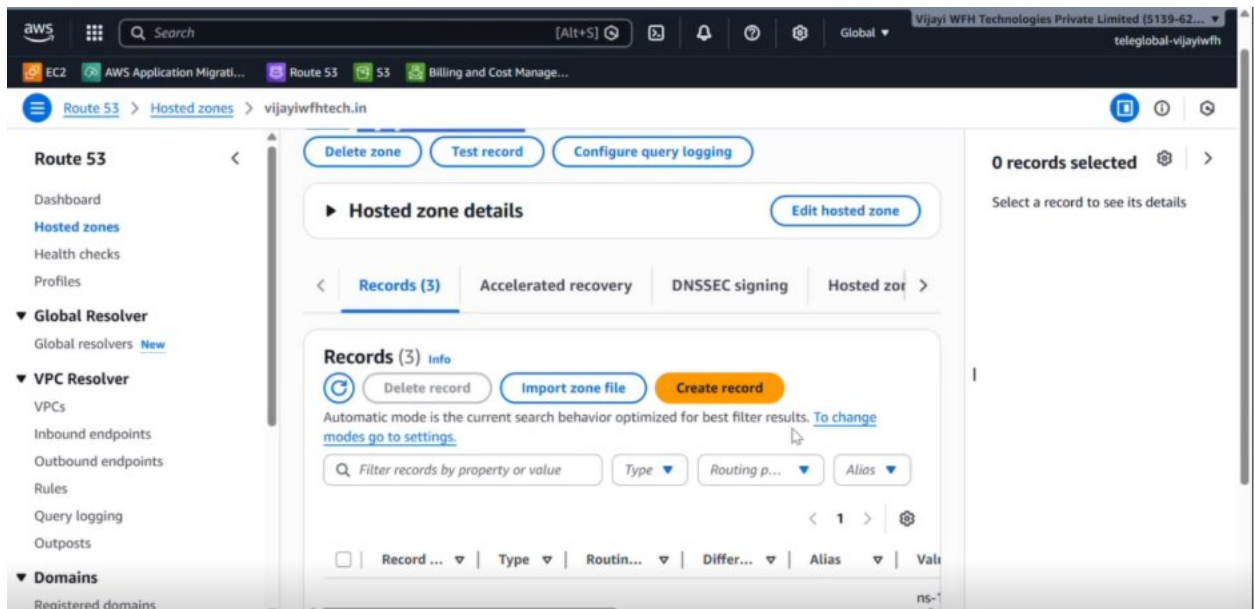
3. Client confirms once the update is completed.

Note: DNS propagation may take up to 24–48 hours depending on the registrar and TTL settings.

Create DNS Record to Route Traffic

Create Record in Hosted Zone

1. Open the hosted zone in Route 53.



2. Click **Create record**.

Quick create record

Record 1

Record name: subdomain (vjayiwhftech.in)

Record type: A - Routes traffic to an IPv4 address and some AWS resources

Alias: ☒ Alias

Value: 192.0.2.235

TTL (seconds): 300 (1m, 1h, 1d)

Routing policy: Simple routing

3. Enter the **Record name** (for example: root domain or subdomain).
4. Select **Record type**: A – IPv4 address.
5. Enable **Alias**.
6. Under **Route traffic to**, select:
 - a. **Alias to Application and Classic Load Balancer**.
7. Select the correct **Region**.
8. Choose the target **Application Load Balancer (ALB)** from the list.

Alias

Route traffic to: Alias to Application and Classic Load Balancer

Region: US East (Ohio)

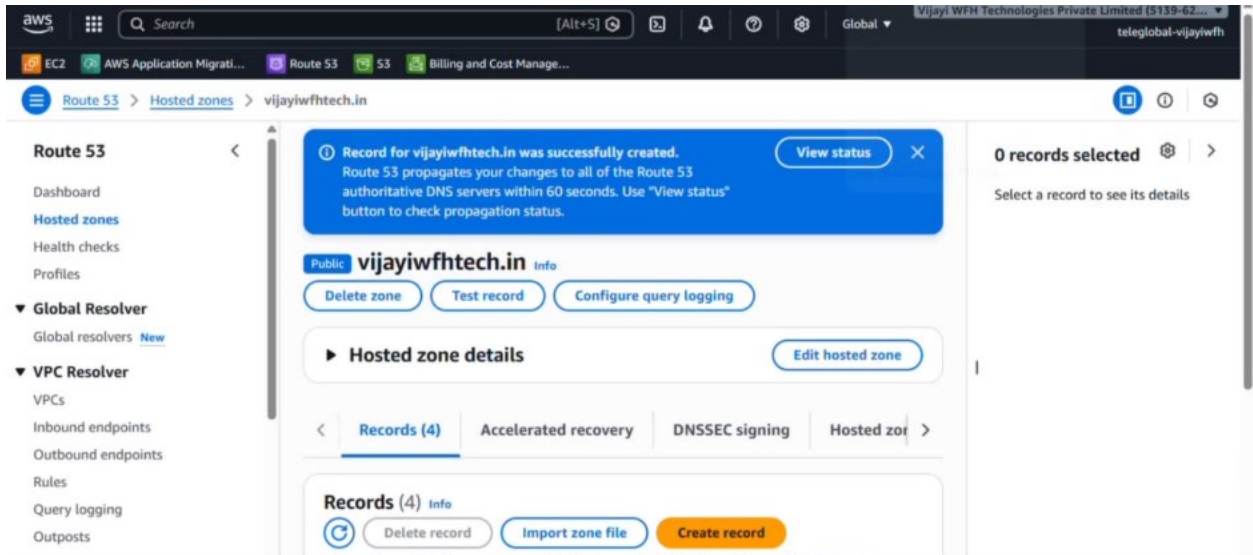
Target: dualstack.vjayiWFH-ALB-555774092.us-east-2.elb.amazonaws.com

Routing policy: Simple routing

Evaluate target health: Yes

Buttons: Add another record, Cancel, Create records

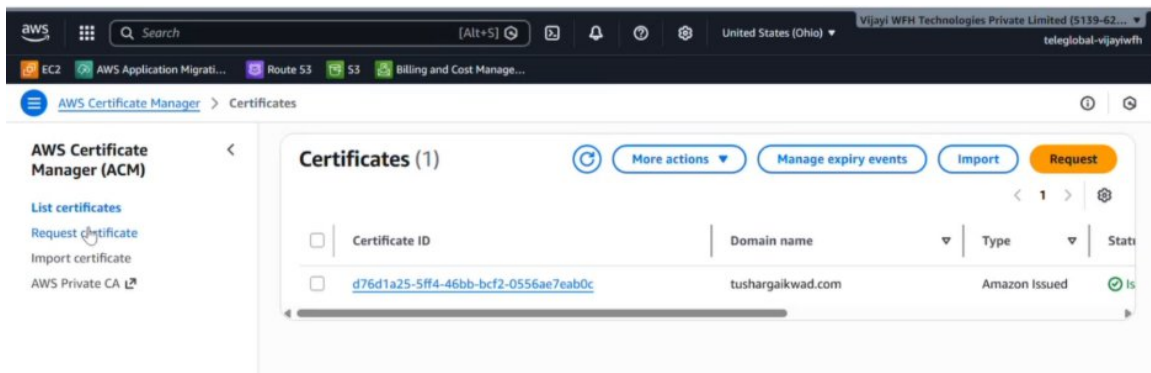
9. Keep routing policy as **Simple** unless otherwise required.
10. Click **Create records**.



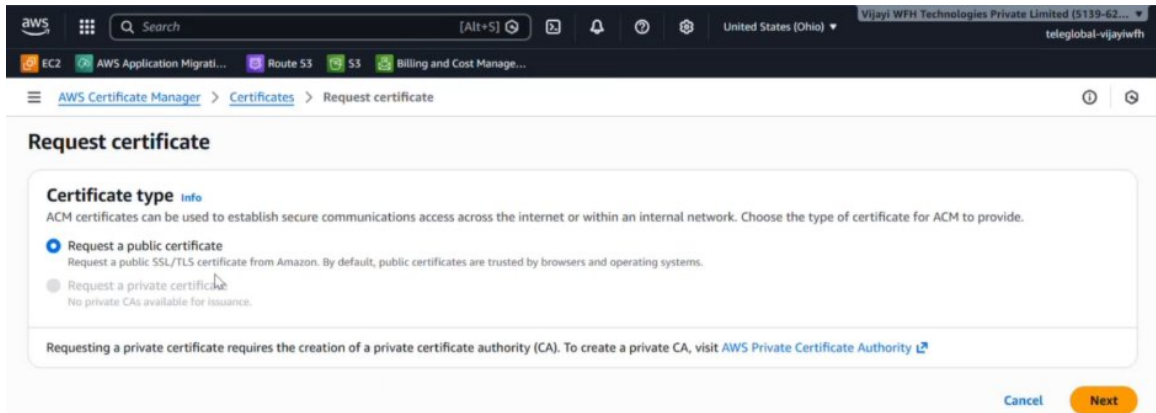
Procedure to Request Certificate in AWS

Access AWS Certificate Manager

1. Log in to the AWS Management Console.
2. In the search bar, type **AWS Certificate Manager** and open the service.
3. Navigate to **Certificates**.



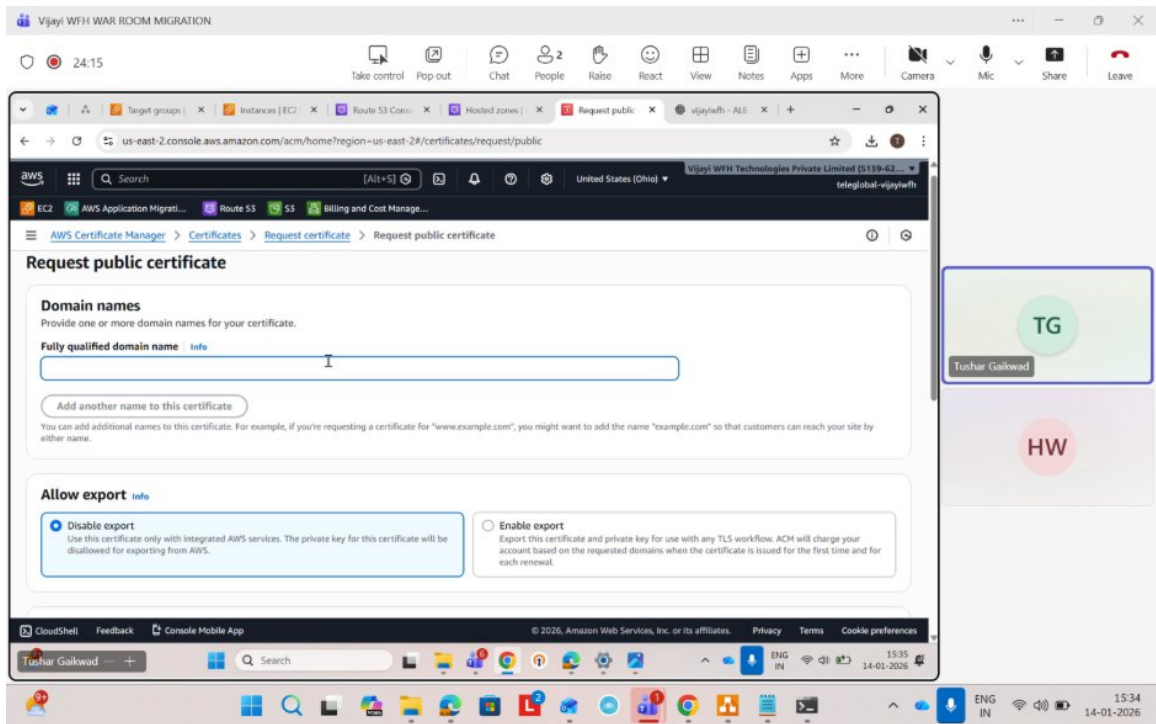
4. Click **Request certificate**.



5. Click Next

Enter Domain Details

1. Select **Request a public certificate** and click **Next**.



2. Enter the **Fully Qualified Domain Name (FQDN)**:
 - a. vijayi-wfh-tech.com

Request public certificate

Domain names
Provide one or more domain names for your certificate.

Fully qualified domain name [Info](#)

[Add another name to this certificate](#)

You can add additional names to this certificate. For example, if you're requesting a certificate for "www.example.com", you might want to add the name "example.com" so that customers can reach your site by either name.

Allow export [Info](#)

☒ **Disable export**
Use this certificate only with integrated AWS services. The private key for this certificate will be disallowed for exporting from AWS.

☐ **Enable export**
Export this certificate and private key for use with any TLS workflow. ACM will charge your account based on the requested domains when the certificate is issued for the first time and for each renewal.

Key algorithm [Info](#)

Select an encryption algorithm. Some algorithms may not be supported by all AWS services.

☒ **RSA 2048**
RSA is the most widely used key type.

☐ **ECDSA P 256**
Equivalent in cryptographic strength to RSA 3072.

☐ **ECDSA P 384**
Equivalent in cryptographic strength to RSA 7680.

Tags [Info](#)

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 tags.

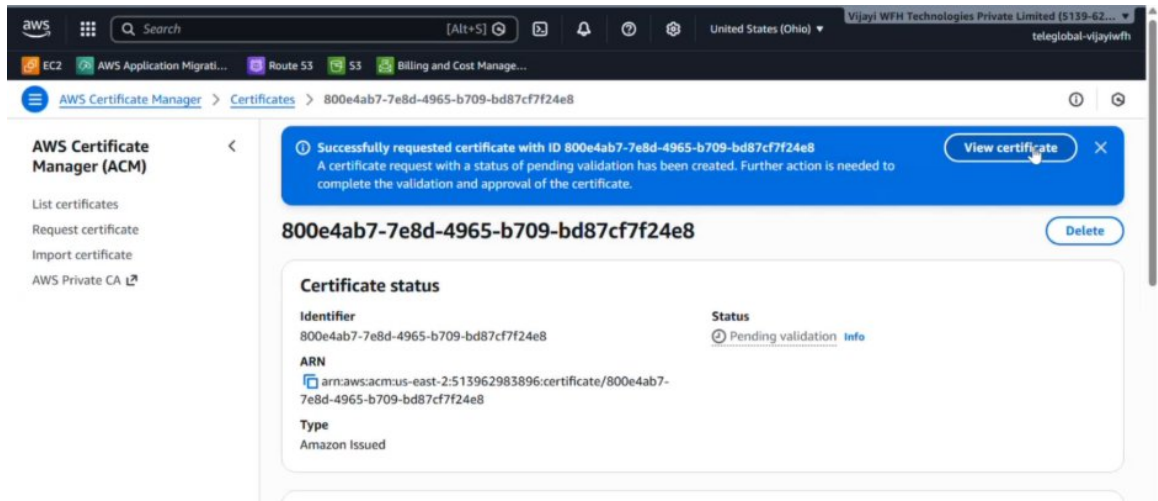
[Cancel](#) [Previous](#) [Request](#)

==

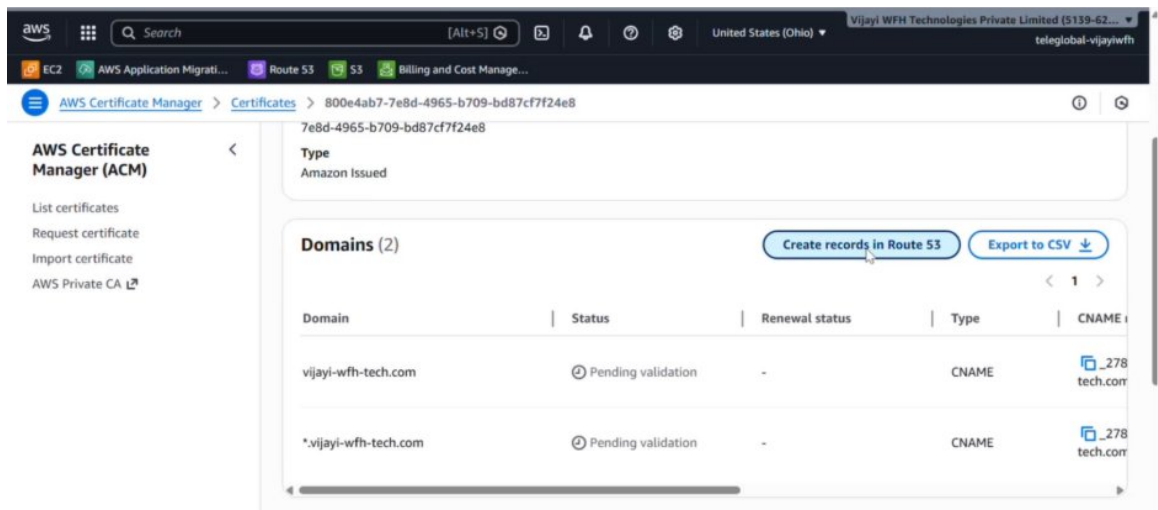
3. Click **Add another name to this certificate**.
4. Enter the wildcard domain:
 - a. *.vijayi-wfh-tech.com
5. Keep all other options as **Default**.
6. Click **Request** to submit the certificate request.

Create DNS Validation Records in Route 53

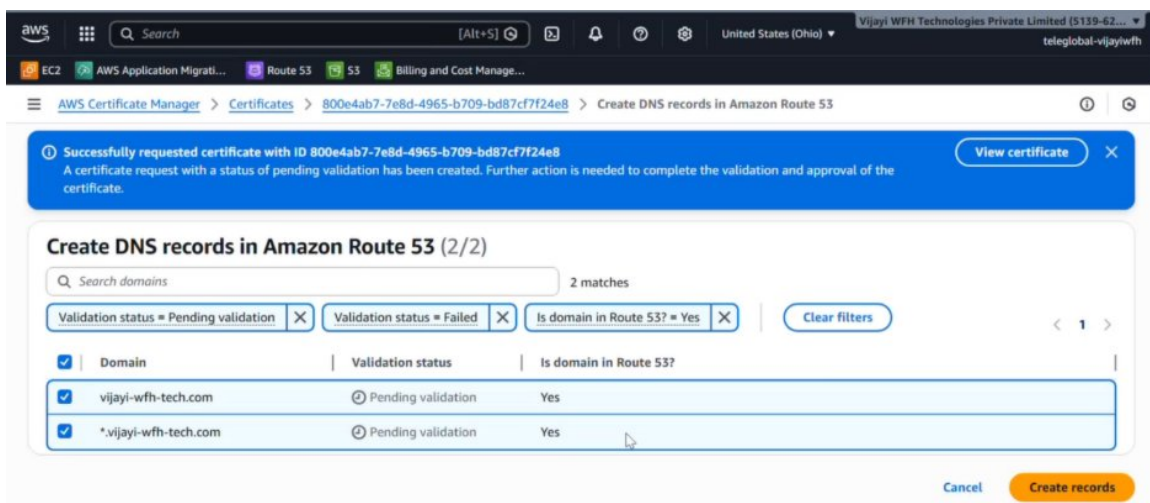
1. After submitting the request, open the newly created certificate.



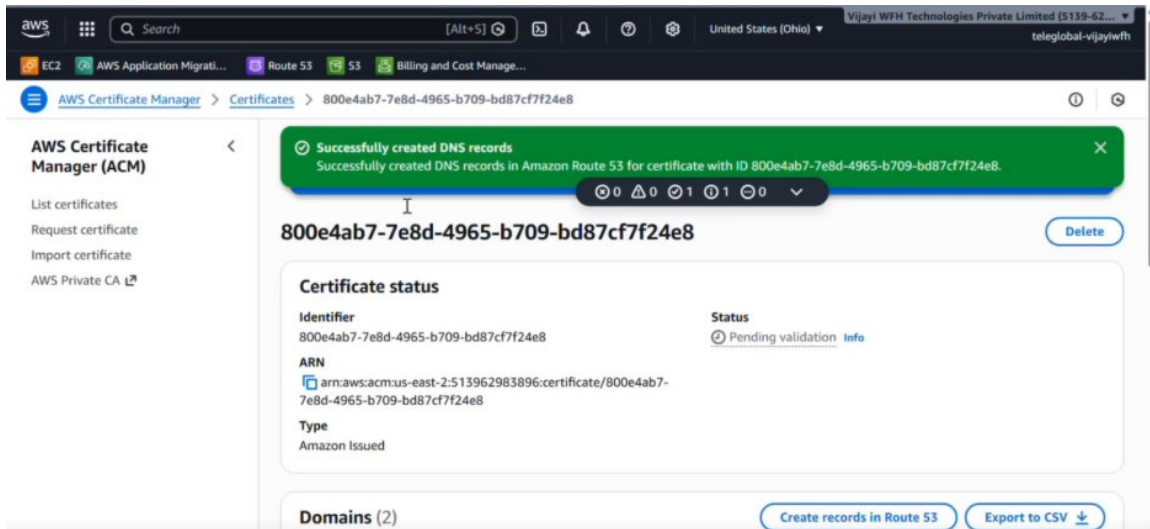
2. Under **Domains**, click **Create records in Route 53**.



3. Review the DNS validation records.

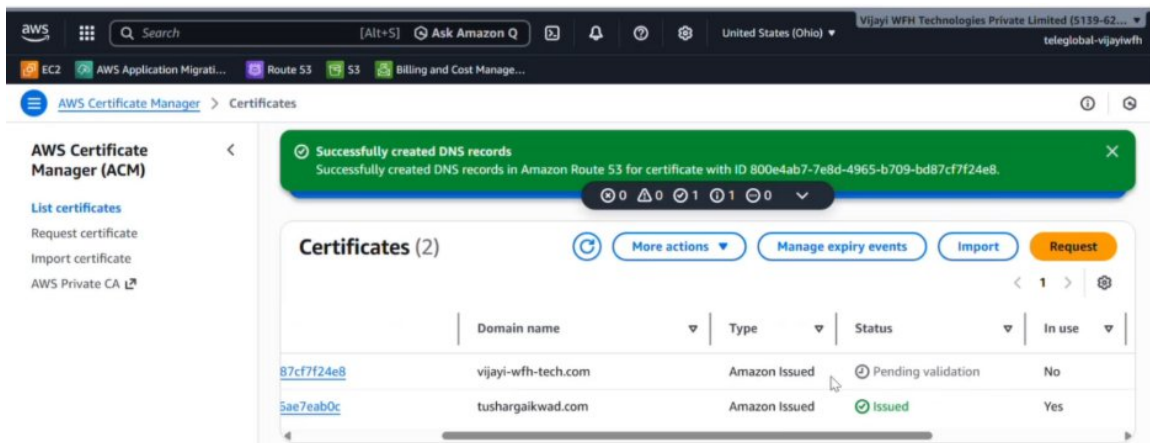


4. Click **Create records** to automatically add validation records to Route 53.

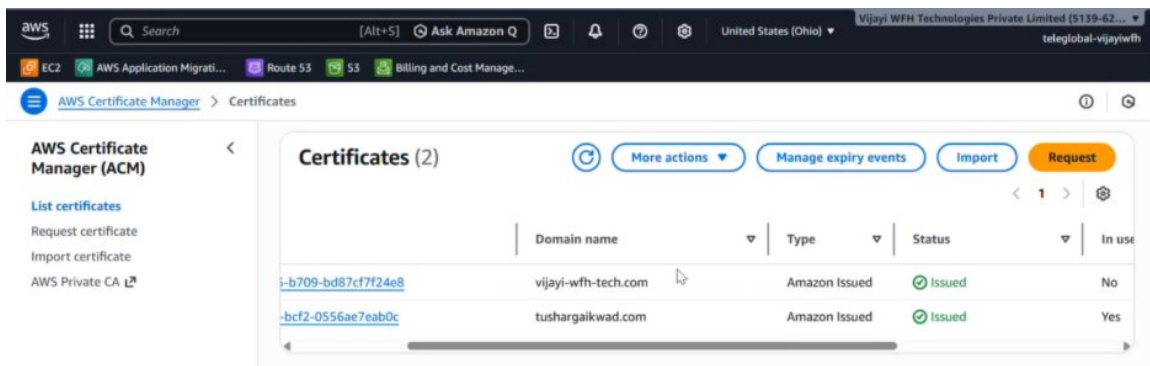


Verify Certificate Status

1. Return to the **Certificates** page in AWS Certificate Manager.
2. Click on the requested certificate.
3. Monitor the **Status** field.

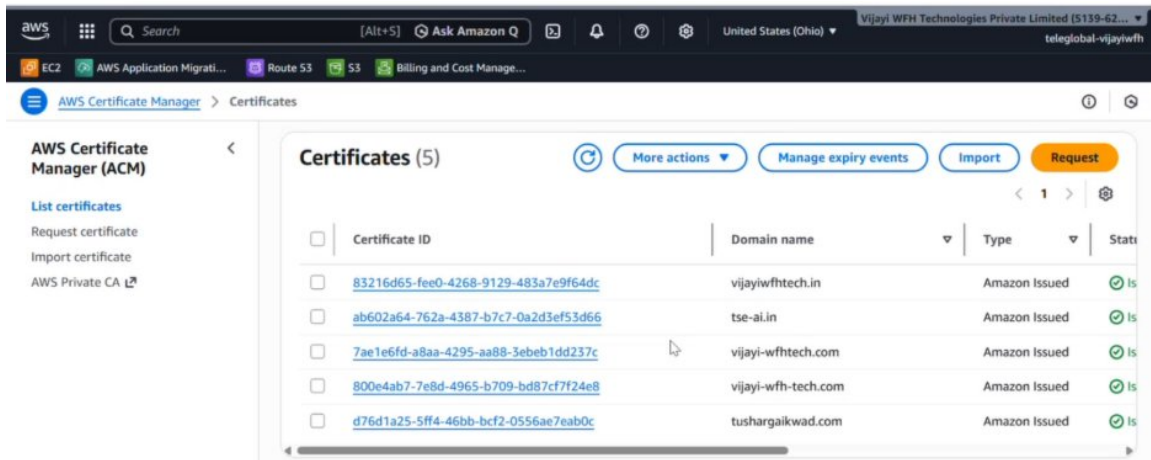


4. Ensure the status changes to **Issued** before proceeding.



Apply Certificate Process for All Hosted Zones

1. Repeat all the steps for each additional hosted zone, if multiple domains are managed.
2. Ensure that all required domains and subdomains have valid issued certificates.



Procedure to Create Target Group for Application Load Balancer

Purpose

This section defines the standard procedure to create a target group in Amazon EC2 for registering instances that will receive traffic from an Application Load Balancer (ALB).

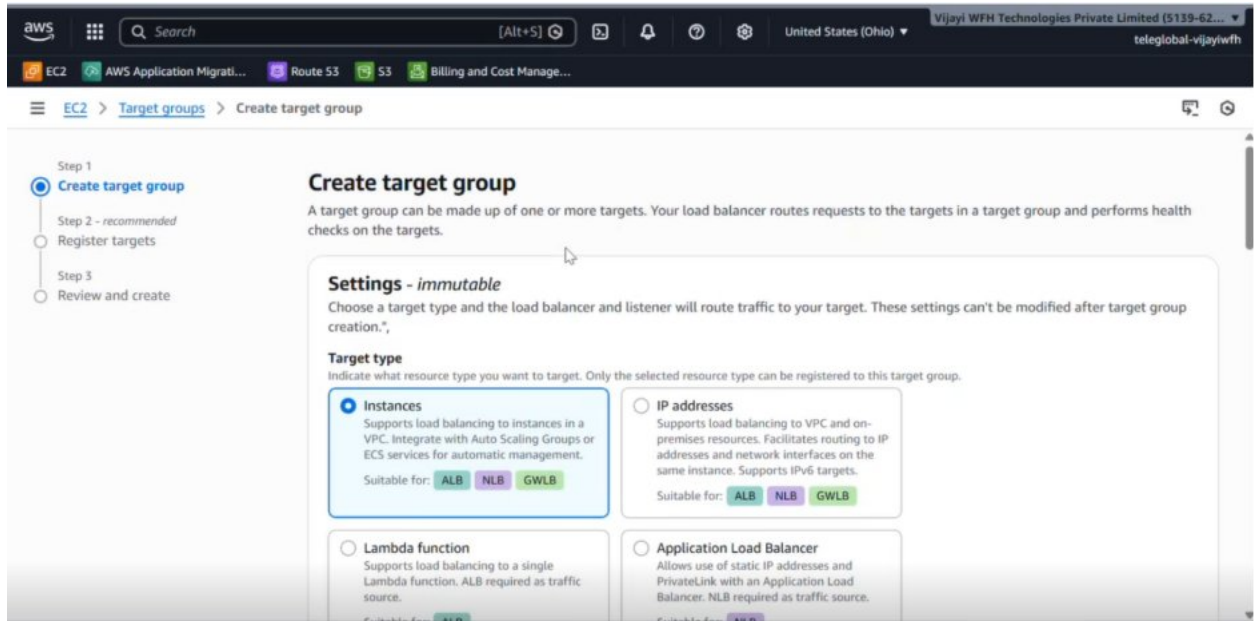
Prerequisites

- Access to AWS Management Console with EC2 and ELB permissions.
- Application Load Balancer has already created.
- Target EC2 instances available in a running state.

Create Target Group

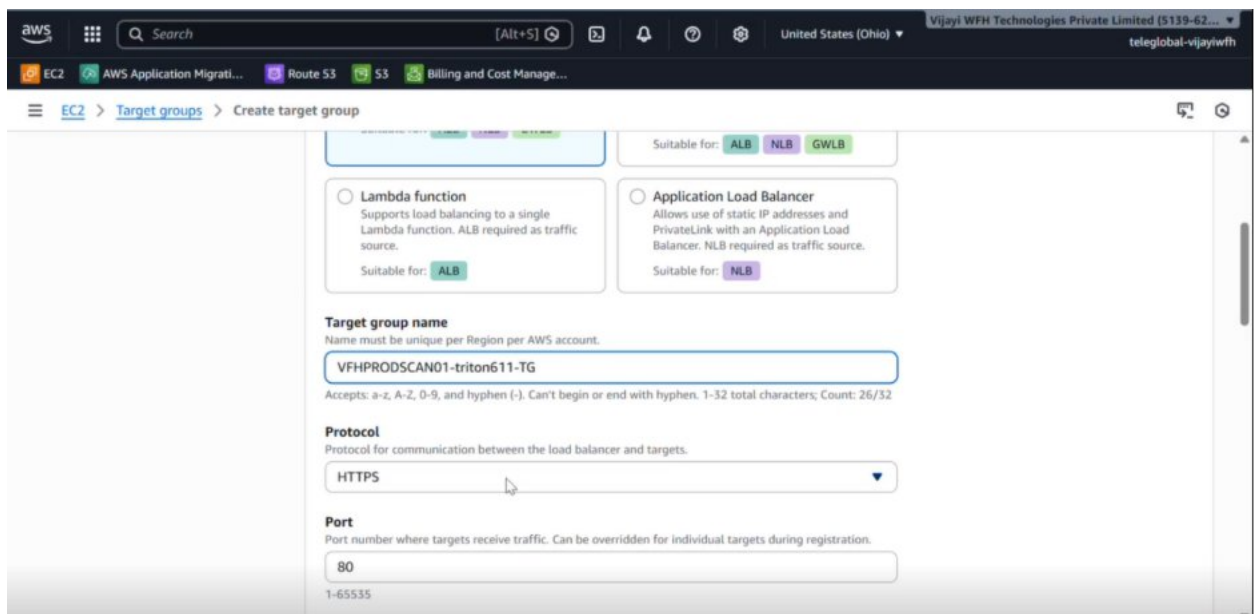
Access EC2 Target Groups

1. Log in to the AWS Management Console.
2. In the search bar, type **EC2** and open the service.
3. In the left navigation pane, under **Load Balancing**, select **Target Groups**.
4. Click **Create the target group**.



Configure Target Group Settings

1. Enter the **Target group name** as per naming standards.



2. Select **Target type**: Instance.

3. Select the appropriate **Protocol** and **Port** (for example: HTTP : 80 or HTTPS : 443).
4. Select the correct **VPC**.
5. Configure **Health check** settings as per application requirements.
6. Click **Next** to proceed.

Target optimizer - optional [Info](#)
Use a target control port when the target has a strict concurrency limit.

Target control port
The port on which the target communicates its capacity. This value can't be modified after target group creation.

Valid range: 1-65535

Attributes

[i](#) Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.

Tags - optional
Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

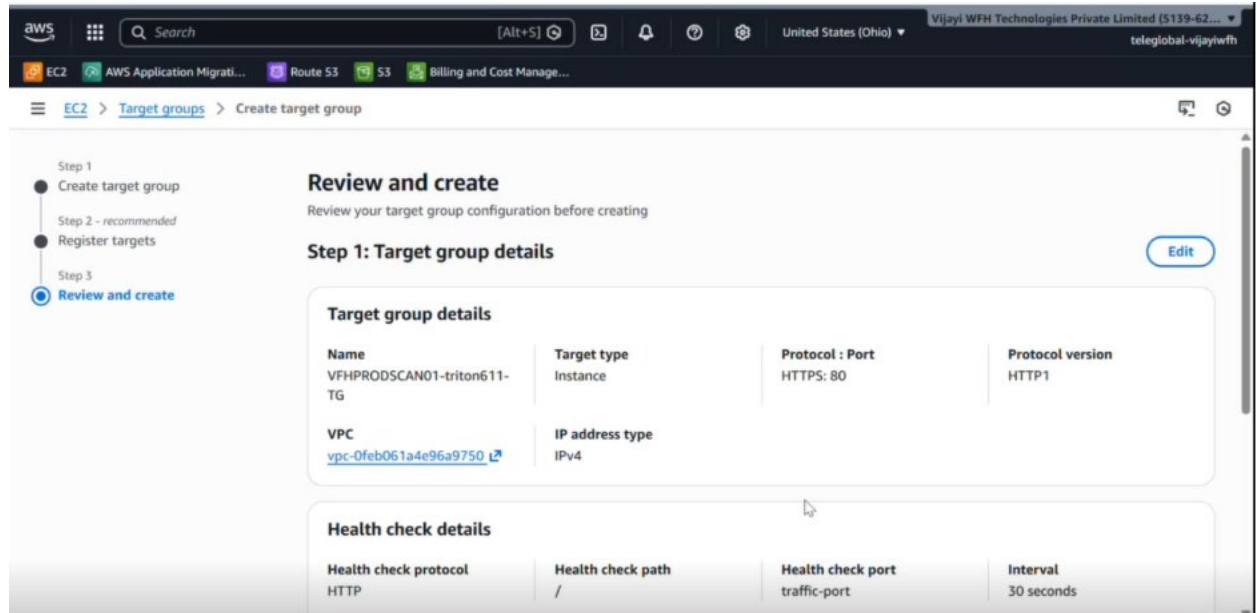
[Cancel](#) [Next](#)

Register Targets

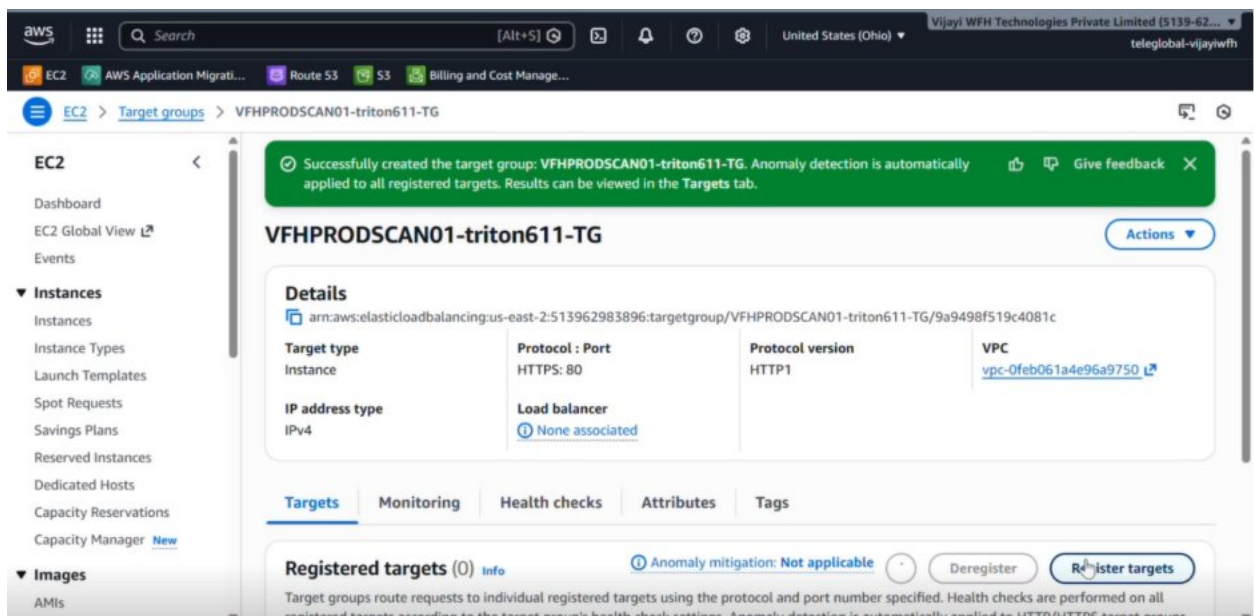
1. From the list of available instances, select the required **Instance ID(s)**.

☐	Instance ID	Name	State	Security groups
☐	i-06356a265b9a99135	VFHDEVMEET01 (holden1107)	Running	VFHDEVMEET01 (holden1107)
☐	i-056e258909cd6d6f0	VFHQAAPP01 (astra1817)	Running	VFHQAAPP01 (astra1817)
☐	i-0c1a1b93fca05c75d	VFHPRODMEET01 (condor3253)	Running	VFHPRODMEET01 (condor3253)
☐	i-04ab8c5c07682c496	VFHPRODSCAN01 (triton611)	Running	VFHPRODMEET01 (triton611)
☐	i-0e378d5aa2dd1343c	VFHPRODAI02 (astra1752) - CL...	Running	VFHPRODAI02 (astra1752)
☐	i-05826273b3d1e09f2	VFHQADB01 (astra1817) - Clone	Running	VFHQADB01 (astra1817)
☐	i-042e3bec65841ecb7	VFHPRODDB01 (astra7187) - ...	Running	VFHPRODDB01 (astra7187)
☐	i-030b3744f0c13b3a8	VFHPRODAPP01 (astra7187)	Running	VFHPRODAPP01 (astra7187)
☐	i-0a6f96516923ac844	VFHDEVAPP01 (astra1802)	Running	VWFHDEVAPP01 (astra1802)
☐	i-09f36cf2d85780c44	OpenVPN-Server	Running	OpenVPN - SG

2. Click **Include as pending**.
3. Review the selected targets in the **Pending** section.

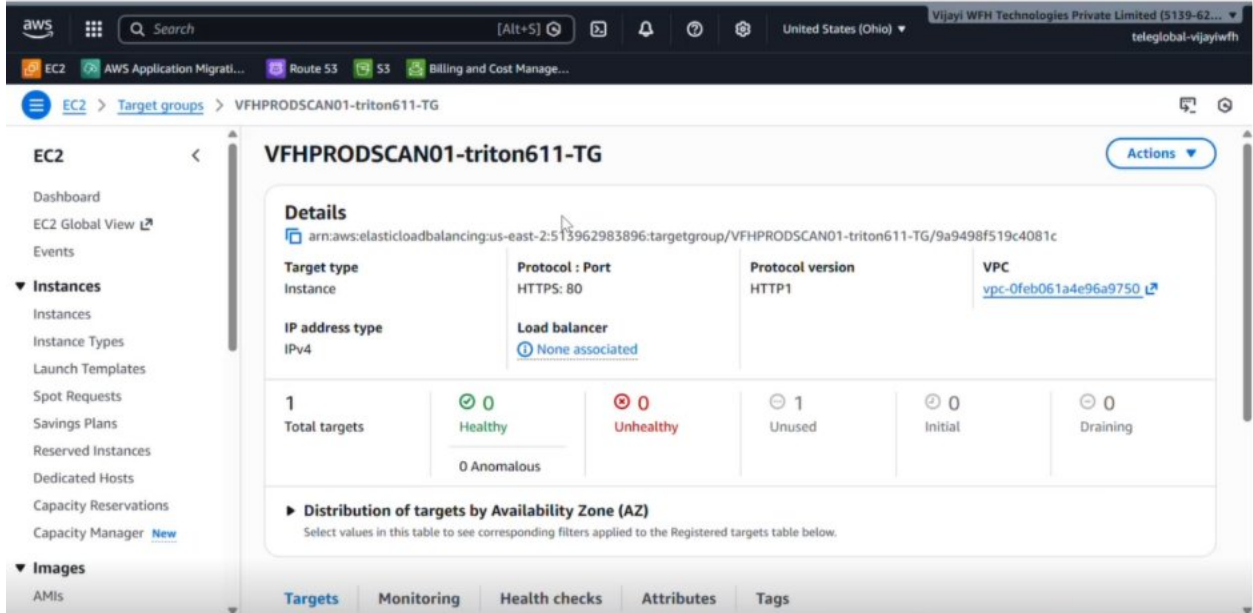


- 4.
5. Click **Create target group** to complete the process.



Post-Creation Validation

1. Open the newly created target group.
2. Verify that the registered instances appear in the **Targets** tab.
3. Confirm the **Health status** changes to **Healthy** after initial checks.



Verify Routing Configuration on Servers

Review Nginx Configuration Files

Routing and virtual host configuration is maintained in the Nginx configuration files on each server.

Configuration Locations:

- /etc/nginx/sites-available/
 - Contains available virtual host configuration files (typically 3 configuration files).
- /etc/nginx/sites-enabled/
 - Contains enabled virtual host configuration files (symbolic links).

Validation Steps

1. Log in to the target server.
2. Navigate to the configuration directories:

```
cd /etc/nginx/sites-available/
ls
```

```
cd /etc/nginx/sites-enabled/  
ls
```

3. Verify that:
 - a. The correct domain names are configured in the server blocks.
 - b. The SSL certificate paths are correctly referenced.
 - c. The required configuration files are enabled.
4. Reload Nginx to apply changes, if required:

```
sudo nginx -t  
sudo systemctl reload nginx
```

Procedure to Configure Listener Rules and SSL Certificate on Application Load Balancer

Purpose

This section defines the standard procedure to configure listener rules on an existing Application Load Balancer (ALB), attach an SSL certificate to the HTTPS listener, and validate secure domain access.

Prerequisites

- Application Load Balancer already created (example: vijayiWFH-ALB).
- Target groups already created and healthy.
- SSL certificate issued and available in AWS Certificate Manager.
- Domain name already mapped to the ALB via Route 53.

Access Load Balancer

1. Log in to the AWS Management Console.
2. In the search bar, type **Load Balancers** and open the service.
3. Select the required load balancer: vijayiWFH-ALB.

Add Listener Rule on HTTPS (Port 443)

Open HTTPS Listener

1. Click on the **Listeners** tab.
2. Select **HTTPS : 443** (for example: 3 rules configured).
3. Click **View/edit rules**.
4. Click **Add rule**.

Configure Host Header Condition

1. Click **Add condition**.
2. Select **Host header**.
3. Enter the required **Domain name** (for example: vijayi-wfh-tech.com).

Configure Target Group Action

1. Under **Actions**, select **Forward to**.
2. Select the appropriate **Target group** from the list.
3. Click **Next**.

Set Rule Priority and Create Rule

1. Enter the **Priority** value as per rule order standards.
2. Click **Next**.
3. Review the rule configuration.
4. Click **Add rule** to save the rule.

Attach SSL Certificate to HTTPS Listener

1. Go back to the **Load balancer** details page.
2. Click on **Listeners** and select **HTTPS : 443**.
3. Click on **Certificates**.
4. Click **Add certificates**.
5. Select the required **Certificate ID** from the list.
6. Click **Include as pending**.
7. Click **Add pending certificates** to apply.

Validate HTTPS and Certificate

1. Open the domain URL in **Incognito mode**.
2. Verify that the site loads over **HTTPS**.
3. Click on the **Lock icon** in the browser address bar.
4. Select **Connection is secure**.
5. Verify that:
 - a. The certificate is valid.

- b. The certificate is issued for the correct domain.

Add Additional Listener Rule (If Required)

1. In the **HTTPS : 443** listener, click **Add rule** again.
2. Configure condition (if required) or leave default.
3. Select the appropriate **Target group**.
4. Click **Next**.
5. Set **Priority** and click **Next**.
6. Click **Add rule** to save.

=====

CloudWatch Alarm

1) Create IAM role for CloudWatch agent

Go to:

IAM → Roles → Create Role

Attach IAM Role with policy:

- CloudWatchAgentServerPolicy
- AmazonSSMManagedInstanceCore

Role Name: EC2-CloudWatchAgent-Role

EC2-CloudWatchAgent-Role info

Allows EC2 instances to call AWS services on your behalf.

Summary

Creation date: February 11, 2026, 12:17 (UTC+05:30)

Last activity: 1 hour ago

ARN: arn:aws:iam::449809970797:role/EC2-CloudWatchAgent-Role

Maximum session duration: 1 hour

Instance profile ARN: arn:aws:iam::449809970797:instance-profile/EC2-CloudWatchAgent-Role

Permissions | Trust relationships | Tags | Last Accessed | Revoke sessions

Permissions policies (2) info

You can attach up to 10 managed policies.

Policy name	Type	Attached entities
AmazonSSMManagedInstanceCore	AWS managed	1
CloudWatchAgentServerPolicy	AWS managed	1

Create Role.

2) Attache IAM role to EC2 instance

Go to:

EC2 → Instances → Select Instance

Then:

Actions → Security → Modify IAM Role

instances (1/1) info

Find Instance by attribute or tag (case-sensitive)

Instance state: running

✓	Name	Instance ID	Instance state	Instance type	Status check	Alarm
	CWAgent-ubuntu	i-Oc80a905a9feb8618	Running	t3.micro		

Actions menu:

- Instance diagnostics
- Instance settings
- Networking
- Security**
- Image and templates
- Storage
- Monitor and troubleshoot

Select: EC2-CloudWatchAgent-Role

Modify IAM role info

Attach an IAM role to your instance.

Instance ID: i-Oc80a905a9feb8618 (CWAgent-ubuntu)

IAM role

Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

EC2-CloudWatchAgent-Role

Buttons: Cancel, Update IAM role

Update IAM rule.

3) Connect to EC2 instance

Instances (1/1) [Info](#)

Find Instance by attribute or tag (case-sensitive) [All states](#)

Instance state = running [Clear filters](#)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
CWAgent-ubuntu	i-0c80a905a9feb8618	Running	t3.micro	3/3 checks passed	View alarms	ap-south-1b	ec2-

i-0c80a905a9feb8618 (CWAgent-ubuntu)

IP name: ip-172-31-2-155.ap-south-1.compute.internal Answer private resource DNS name: IPv4 (A) Auto-assigned IP address: 3.7.69.140 [Public IP]	Instance type: t3.micro VPC ID: vpc-06dffc556fb2e4d59 Subnet ID: subnet-0f49fe24fa82852c0 Instance ARN: arn:aws:ec2:south-1:449809970797:instance/i-0c80a905a9feb8618	Elastic IP addresses: - AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendations. Auto Scaling Group name: - Managed: false
---	--	--

For **Amazon Linux 2**:

- **Update System Packages**

```
sudo yum update -y
```

```
sudo yum install amazon-cloudwatch-agent -y
```

- **Check version**

```
/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a status
```

- **Download CloudWatch Agent package for Ubuntu**

```
wget https://s3.amazonaws.com/amazoncloudwatch-agent/ubuntu/amd64/latest/amazon-cloudwatch-agent.deb
```

- **Install the package**

```
sudo dpkg -i -E ./amazon-cloudwatch-agent.deb
```

- **Fix dependencies (if required)**

```
sudo apt --fix-broken install -y
```

- **Check CloudWatch Agent Status**

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -  
a status
```

- **Start CloudWatch Agent Using Config (Ubuntu)**

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \  
-a fetch-config \  
-m ec2 \  
-c file:/opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-  
agent.json \  
-s
```

Config file:

```
{  
  "agent": {  
    "metrics_collection_interval": 60,  
    "run_as_user": "root"  
  },  
  "metrics": {  
    "namespace": "CWAgent",  
    "append_dimensions": {  
      "InstanceId": "${aws:InstanceId}",  
      "InstanceType": "${aws:InstanceType}"  
    },  
    "metrics_collected": {  
      "mem": {  
        "measurement": [  
          "mem_used_percent"  
        ],  
        "metrics_collection_interval": 60
```

```

    "metrics_collection_interval": 60
  },
  "disk": {
    "measurement": [
      "used_percent"
    ],
    "metrics_collection_interval": 60,
    "resources": [
      "/"
    ]
  }
}
}
}
}

```

- CTRL + O
- Enter
- CTRL + X

- **Enable on boot:**

```

sudo systemctl enable amazon-cloudwatch-agent
sudo systemctl restart amazon-cloudwatch-agent

```

- **Check status:**

```

sudo systemctl status amazon-cloudwatch-agent

```

Confirm Metrics Are Coming in CloudWatch

Go to:

CloudWatch → Metrics → All metrics → CWAgent

You should see:

disk_used_percent

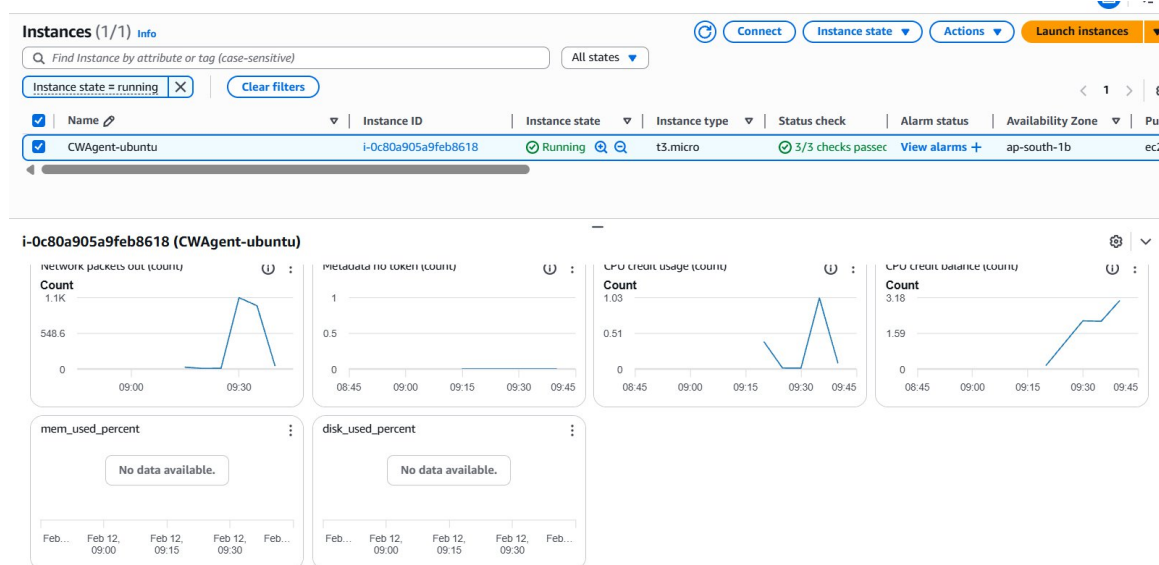
mem_used_percent

Enable CWAgent Metrics in EC2 Monitoring Tab

Go to:

EC2 → Instances → Select Instance → Monitoring tab

Enable toggle:



For Amazon Linux 2:

`sudo yum update -y`

`sudo yum install amazon-cloudwatch-agent -y`

Check version:

```
/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a status
```

- Create CloudWatch Agent Config (ONLY Required Metrics)

Create config file:

```
sudo vi /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json
```

Paste this CONFIG:

```
{
  "agent": {
    "metrics_collection_interval": 60,
    "run_as_user": "root"
  },
  "metrics": {
    "namespace": "CWAgent",
    "append_dimensions": {
      "InstanceId": "${aws:InstanceId}",
      "InstanceType": "${aws:InstanceType}"
    },
    "metrics_collected": {
      "mem": {
        "measurement": [
          "mem_used_percent"
        ],
        "metrics_collection_interval": 60
      },
      "disk": {
        "measurement": [
          "used_percent"
        ],
        "metrics_collection_interval": 60,
        "resources": [
          "/"
        ]
      }
    }
  }
}
```

```
}
```

Save and exit.

- Start CloudWatch Agent Using This Config

Run:

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
-a fetch-config \
-m ec2 \
-c file:/opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json \
-s
```

Enable agent on boot:

```
sudo systemctl enable amazon-cloudwatch-agent
sudo systemctl restart amazon-cloudwatch-agent
```

Check status:

```
sudo systemctl status amazon-cloudwatch-agent
```

- Confirm Metrics Are Coming in CloudWatch

Go to:

CloudWatch → Metrics → All metrics → CWAgent

You should see:

disk_used_percent

mem_used_percent

- Enable CWAgent Metrics in EC2 Monitoring Tab

Go to:

EC2 → Instances → Select Instance → Monitoring tab

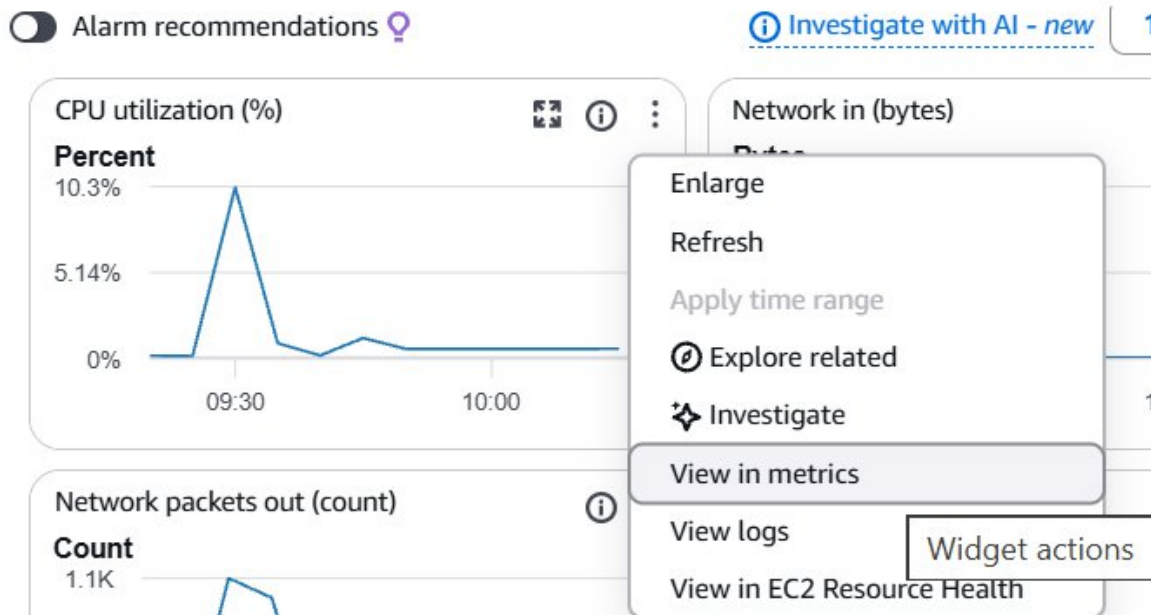
Enable toggle:

Set Alarm for CPU utilization:

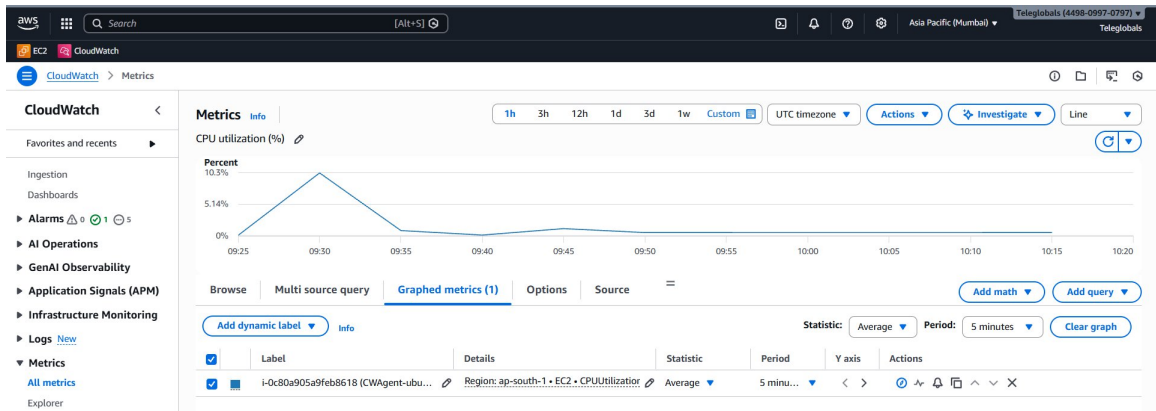
Go to:

EC2 → Monitoring → CPU Utilization → click on triple . → View in metrics

i-0c80a905a9feb8618 (CWAgent-ubuntu)



Next tab will open with cloudwatch



Click on Alarm sign

This screenshot shows the bottom portion of the CloudWatch Metrics console. It includes the 'Add dynamic label' button, the 'Info' link, and the 'Statistic' dropdown set to 'Average'. The 'Period' is set to '5 minutes'. Below the table, there is a 'Create alarm' button.

Create alarm

The screenshot shows the AWS CloudWatch Alarms console. The left sidebar contains navigation links for CloudWatch, Alarms, Create alarm, Alarm recommendations available, Step 1: Specify metric and conditions, Step 2: Configure actions, Step 3: Add alarm details, Step 4: Preview and create, and Alarm recommendations. The main area displays the 'Specify metric and conditions' step. It includes a graph for 'CPU utilization (%)' over a 1-hour period, showing a peak of 10.3% at 09:30. To the right of the graph, there is a form with fields for 'Namespace' (AWS/EC2), 'Metric name' (CPUUtilization), 'Instanced' (i-0c80a905a9feb8618), 'Instance name' (CWAgent-ubuntu), 'Statistic' (Average), and 'Period' (5 minutes). There is an 'Edit' button at the top right of the form.

CloudWatch > Alarms > Create alarm

Statistic
Average

Period
5 minutes

Conditions

Threshold type
☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever CPUUtilization is...
Define the alarm condition.

☒ Greater
> threshold

☐ Greater/Equal
≥ threshold

☐ Lower/Equal
≤ threshold

☐ Lower
< threshold

than...
Define the threshold value.

10000

Must be a number.

► Additional configuration

Cancel Next

Define threshold value and click on Next

Conditions

Threshold type
☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever CPUUtilization is...
Define the alarm condition.

☒ Greater
> threshold

☐ Greater/Equal
≥ threshold

☐ Lower/Equal
≤ threshold

☐ Lower
< threshold

than...
Define the threshold value.

75

Must be a number.

► Additional configuration

Cancel Next

click on create new topic

CloudWatch > Alarms > Create alarm

Alarm recommendations available
Turn on Recommendations to pre-populate the wizard with the recommended alarms.

Step 1
Specify metric and conditions

Step 2
Configure actions

Step 3
Add alarm details

Step 4
Preview and create

Configure actions

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ In alarm
The metric or expression is outside of the defined threshold.

☐ OK
The metric or expression is within the defined threshold.

☐ Insufficient data
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☐ Select an existing SNS topic

☒ Create new topic

☐ Use topic ARN to notify other accounts

Create a new topic...
The topic name must be unique.

Enter topic name

SNS topic names can contain only alphanumeric characters, hyphens (-) and underscores (_).

Email endpoints that will receive the notification...
Add a comma-separated list of email addresses. Each address will be added as a subscription to the topic above.

user@example.com

user1@example.com, user2@example.com

Create topic

Create new topic:

Create a new topic: name

Email endpoints that will receive the notification...

☐ Select an existing topic

☒ Create new topic

☐ Use topic ARN to notify other accounts

Create a new topic...
The topic name must be unique.

Teleglobal

SNS topic names can contain only alphanumeric characters, hyphens (-) and underscores (_).

Email endpoints that will receive the notification...
Add a comma-separated list of email addresses. Each address will be added as a subscription to the topic above.

harshal.wakde@teleglobals.com

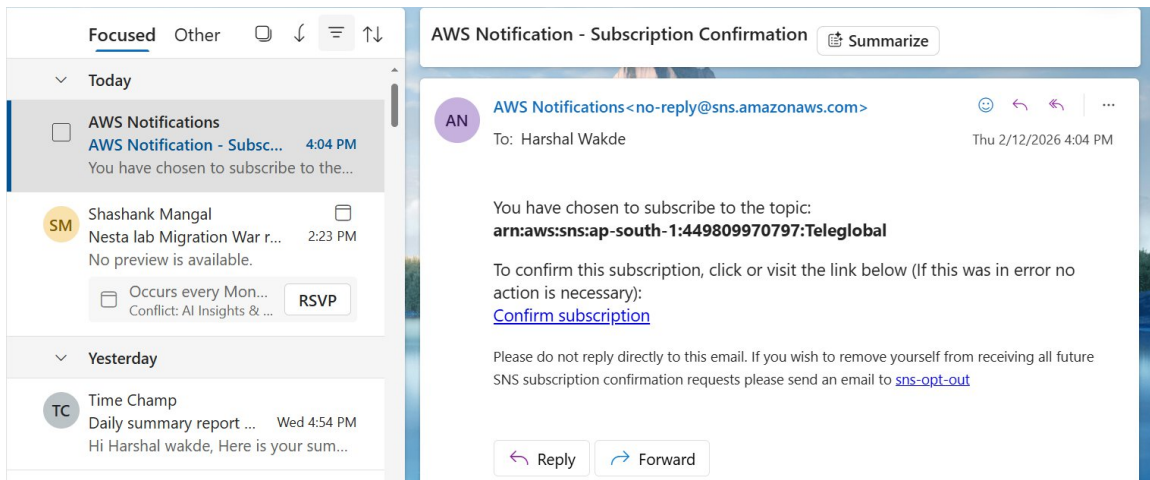
user1@example.com, user2@example.com

Create topic

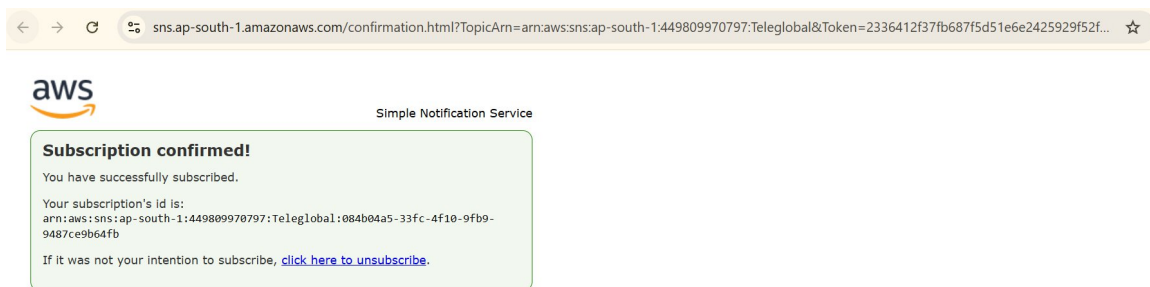
Add notification

Click on Create topic → Next

Mail will come



Click on confirm subscription



Alarm name:

Alarm details:

Next

EC2CloudWatch

CloudWatchAlarmsCreate alarm

Alarm recommendations available
Turn on Recommendations to pre-populate the wizard with the recommended alarms.

Step 1
Specify metric and conditions

Step 2
Configure actions

Step 3
Add alarm details

Step 4
Preview and create

Add alarm details

Name and description

Alarm name
PROD Server CPUUtilization>=75

Alarm description - optional [View formatting guidelines](#)

Edit

Preview

PROD Server CPUUtilization >= 75%

Up to 1024 characters (33/1024).

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Tags - optional [Info](#)

No tags associated with the resource.

Add new tag

You can add up to 50 tags.

Cancel

Previous

Next

EC2CloudWatch

CloudWatchAlarmsCreate alarm

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Turn on Recommendations to pre-populate the wizard with the recommended alarms.

Step 1
Specify metric and conditions

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Preview and create

Step 1: Specify metric and conditions

Metric

Graph
This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.

Percent
82.5%

75

41.3%

0%

08:0008:3009:0009:3010:0010:30

i-0c80a905a9feb8618 (CWAgent-ubuntu)

Namespace
AWS/EC2

Metric name
CPUUtilization

Instanceid
i-0c80a905a9feb8618

Instance name
CWAgent-ubuntu

Statistic
Average

Period
5 minutes

Conditions

Threshold type
Static

Whenever CPUUtilization is
Greater (>)
than...
75

Additional configuration

Step 2: Configure actions

Actions

Notification
When in alarm, send a notification to "Telglobal"

Step 3: Add alarm details

Alarm details

Name
PROD Server CPUUtilization>=75

Description
PROD Server CPUUtilization >= 75%

Tags (0)

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Cancel

Previous

Create alarm

create alarm

Successfully created alarm PROD Server CPUUtilization>=75.

View alarm

Alarms (7)

Q Search

Alarm state: Any

Alarm type: Any

Actions status: Any

< 1 >

Hide Auto Scaling alarms

Clear selection

Create composite alarm

Actions

Create alarm

	Name	State	Actions	Last state update (UTC)	Conditions
☐	PROD Server CPUUtilization>=75	Insufficient data	Actions enabled	2026-02-12 10:43:37	CPUUtilization > 75 for 1 datapoints within 5 minutes

Follow the same process to set alarm for disk uti, mem uti and etc.